

Role Differentiation as Ignition of a Collective Information Engine: A Theory of Structuration in Agent Populations

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Informational active matter has shown how measurement-informed decisions produce collective order, so far in systems that reach consensus. We design a collective information engine that orders by differentiation instead, and construct a minimal instance using anti-coordination games, the simplest game class in which differentiated role information has value. Agents infer their role from a noisy social signal grounded in a persistent identity, and role-following action feeds back into that signal, which shapes the incentive to follow roles. Resources accrued through coordinated role-play combine with identity variability to reinforce the schemas that generated them. The model thereby operationalizes Sewell’s duality of schemas and resources in Structuration, the resolution of sociology’s structure–agency debate. The engine ignites when a social loop gain—the product of identity persistence, cognitive capacity, channel fidelity, and schema strength—exceeds one. For a repertoire of such schemas, roles emerge with increasing gain in a bifurcation cascade whose functional form is fixed by the repertoire’s eigenvalue spectrum, ranging from monitorable logarithmic sequences to avalanches that arrive without warning. Resource accumulation supplies the fitness of a replicator dynamics on schema strengths, which selects the cascade type endogenously. Subcritical behavioral covariance reveals that type before onset, enabling early detection, while feedback channel parameters bias which type is selected. Platform design then becomes a control lever to throttle emergent coordination. This theory grounds distributional AGI takeoff in a mechanism and provides a monitor-based solution. Joining game theory, collective dynamics, and information engines, we open a route to an information thermodynamics of agent populations.

INTRODUCTION

A century-old question runs through the social sciences: how does macro-level social structure shape individuals while, simultaneously, individuals’ choices generate that structure? Giddens proposed *structuration* as a resolution to this *structure–agency debate* based on the *duality of structure* [1]: social structure is both the medium and the outcome of the schemas and resources it organizes. Soon after, another sociologist, Sewell, noted that Giddens’s duality accounted only for the reproduction of schemas—essentially a local stability condition rather than an endogenous dynamics. Motivated to provide a qualitative account of schema transformation, Sewell recast the duality as a feedback process between *schemas*, the transposable rules and conventions that give meaning to resources by prescribing behavior around their use, and *resources*, the material and symbolic stocks that schemas mobilize and that empower the agents using them [2]. Each sustains the other: schemas direct the use of resources, and resource-wielding agents reproduce (or transform) schemas. Debates resonant with the same micro–macro tension, many unresolved, recur across social science (*e.g.* in cultural evolution [3] and in macroeconomics [4, 5]). An explicit dynamics of structuration would help ground those debates in quantitative models.

Despite half a century of agent modeling, Sewell’s account of resource-schema structuration has no quantitative formalization. Existing agent modeling frameworks

do not operationalize its core objects. Opinion dynamics has one state variable per agent and no resource that accumulates. Cultural evolution uses transmission and replicator models (*e.g.* evolutionary game theory, including spatial reaction-diffusion extensions) that take the trait space or the game as given, without resource accumulation. Agent-based computational economics does accumulate resources under institutions [6], but its institutions are rules imposed on the agents rather than objects the dynamics selects. More specialized models treat resource accumulation [7] and payoff-dependent network reshaping [8], but represent schemas as scalar state or payoff terms rather than as semantically loaded, transposable rules. None of these treats the resource-schema dynamics central to Sewell’s vision.

Large language models have opened a new class of applications for agent modeling, both of human social systems [9] and of deployed AI agent systems [10]. Calls for applications of collective dynamics [11] and systems theory [12] now have responses: impressive applications of established frameworks, among them the statistical mechanics of opinion dynamics [13] and replicator-like mean field theories [14]. Structuration, however, has no such framework to apply. Supplying one now would support research on a variety of timely questions about deployed AI, including agent personas and coordination.

We provide such a theory, exemplified for role differentiation in anti-coordination games, that introduces a new kind of informational active matter. Systems that convert measurement into extractable work through feedback [15, 16], subject to performance limits set by measurement quality [17], are known as *information engines*.

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First demonstrated by Szilard almost 100 years ago [18], they are now understood as using information as a thermodynamic resource: measurement makes it available and feedback converts it into work [19, 20]. We show that a population of agents incentivized and provided with the means to role-play via identity signals constitutes a *collective, self-fueling information engine*. Coordinated role-play is the payoff-generating output—the engine’s analog of extracted work—that is bounded by the measured role information, just as Szilard’s bound limits the work extracted by his illustrative model engine. As an instance of the long-noted link between Maxwell’s demon and symmetry-breaking order parameters [21], this description belongs to the emerging field of informational active matter: a modern effort to extend Szilard’s ideas to more complex, in particular many-body settings, where measurement-informed decisions produce collective order bounded by the information acquired [22, 23]. Those engines reach *consensus*, a magnetization-like order. Each measurement here is of another agent, and it informs which role to take. The engine produces *differentiation*, a covariance-like order in which agents take complementary rather than matching roles. Social role structure is the engine’s running state, sustained by the burning of role information generated by that structure.

Modelling many-agent systems with tractable social structure is challenging. Schema transformation in the strict sense requires specifying a mechanism holding schemas on distinct axes. A breadth of choices differ in their assumptions and yield a family of theories (Appendix D). In this paper, we center the structuration that every such theory contains by scoping down onto the dynamics of schema strengths for a fixed repertoire to explain the schema–resource loop. Which schemas become institutionalized, in what order, and which agents are enriched are endogenous outcomes, while what a schema is *about* is held fixed. Our main results are presented in dedicated sections:

- I. a minimal model that couples schemas and resources through agent identities and actions;
- II.A. an ignition condition for self-sustaining role differentiation—an eigenvalue of the social loop gain Λ exceeding one, where Λ factorizes into identity persistence, agent cognitive capacity, social feedback channel fidelity, and the schema repertoire;
- II.B. a self-consistent stationary state, in which the loop closes into a condition on the schema repertoire alone, with resource weighting sustaining a diversity of schemas evoking complex society;
- III. a bifurcation cascade under extension to multiple schema contexts, in which role dimensions activate sequentially, with functional form determined by the eigenvalue spectrum of the schema repertoire;
- IV. an endogenous theory of that spectrum: a replicator dynamics on schema strengths, with fitness the resource-weighted variance and total regulated by a finite signaling capacity, which in the near-threshold mean field selects among cascade classes, from single

avalanches to logarithmic cascades; and

- V. a monitoring protocol—subcritical spectral inversion of the behavioral covariance—that reads out the future cascade of the current repertoire before its onset, subject to finite-sample resolution and slow schema drift.

The first three results formulate a linearization about the role-free state that gives the loop gain Λ , whose eigenvalues set the thresholds for role differentiation, and closed on itself it gives the stationary repertoire. Its spectrum is what III classifies; IV decides which spectrum the loop produces; and V recovers that spectrum from the subcritical covariance before the cascade begins.

We thus demonstrate that two ingredients suffice for roles to emerge: identities that persist and are legible to others—here lossy summaries of context-dependent action histories—and schema selection driven by the resource-weighted covariance of those identities. The role structure they produce is a collectivist solution to the coordination dilemma facing large agent populations [24]: how to organize a role-based society that offers more payoff than the individually rational equilibrium (Sec. I). The discussion application to distributional AGI [25]: the hypothesis that general-level capability may emerge from a coordination phase transition among many individually-limited AI agents. An operator who fixes the environment before phase onset (‘ignition’) holds a control surface over which takeoff scenario is reachable. Finally, we set out what differentiation adds to informational active matter research. We argue that AI agent population systems are a promising experimental system in which structuration loops like the one articulated here can be built and measured in pursuit of developing a quantitative theory of agent populations.

I. A MINIMAL ENGINE OF SOCIAL STRUCTURE

Figure 1a schematizes the model described in this section to illustrate the closed loop: three slower varying state variables coupled through a fast per-encounter chain of status, action and reward. We build it in four steps: the game that makes roles worthwhile, the identities that realize them, the resources that accumulate, and the schemas that survive.

The game. We design utility into roles minimally, using a generic anti-coordination game. Two players choose to Go or Defer: both playing Go is costly to both $(-1, -1)$, both deferring is mediocre $(3, 3)$, and complementary role-play pays best on average, though asymmetrically $(6, 2)$. The latter is played in a correlated equilibrium that pays 4 per player (on average from random assignment), while the mixed Nash equilibrium pays only 2.5. We realize this correlated equilibrium using a role signal as the correlating device. Even though role-following is not a dominant strategy—it is obeyed only when partners also follow—its incentive constraint is strict once

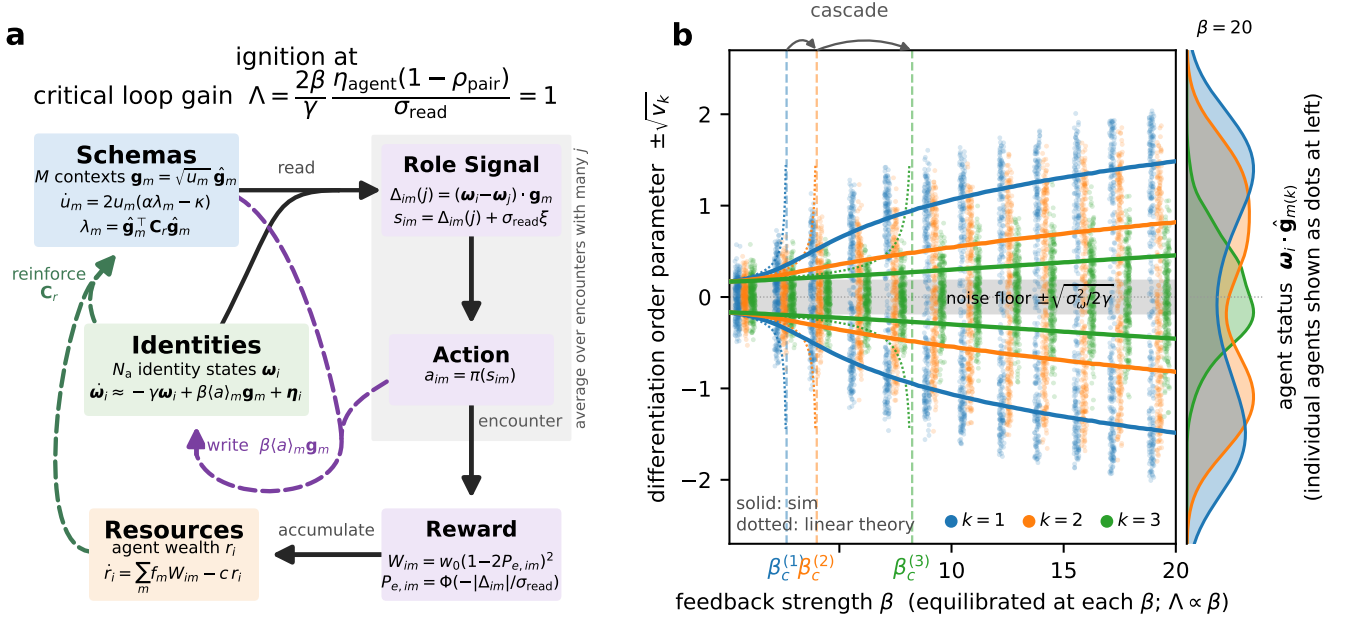


FIG. 1. **The self-consistent schema–role–resource loop and its ignition.** (a) Model schematic, separating the three persistent state variables (schemas $\mathbf{g}_m = \sqrt{u_m} \hat{\mathbf{g}}_m$, of which only u_m evolves; identities $\boldsymbol{\omega}_i$; resources r_i) from the fast per-encounter chain (status, action, reward). Solid arrows are the feed-forward read-out: schema and identity form the status signal, the policy thresholds it into an action, and the coordination reward accumulates as resources. The shaded band is one encounter, averaged over the many partners j drawn across encounters, after which the status difference enters only through Δ_{im} . Dashed arrows are the slow state updates that close the loop: resources reweight identities to reinforce schemas through the resource-weighted covariance $\mathbf{C}_r$, and actions are inscribed back into identities along the same axis (the write $\beta(a)_m \mathbf{g}_m$). The single direction $\mathbf{g}_m$ is thus both read and write axis, which makes Sewell’s schema–role duality a fixed-point loop, and the engine ignites when the loop gain Λ exceeds one (top). (b) Three sequential bifurcations (differentiation order parameter vs. feedback strength) over the β sweep ($N_a = 2500$, $d = M = 3$, hierarchical fixed schema repertoire, random pairing; Appendix B). Points are individual agent statuses $\boldsymbol{\omega}_i \cdot \hat{\mathbf{g}}_{m(k)}$ along the three schema axes, with $m(k)$ the context whose strength gives the k -th eigenvalue. Solid curves are the order parameter $\pm\sqrt{v_k}$, the two branches of a pitchfork on each axis, with v_k the k -th eigenvalue of the population covariance $\mathbf{C}$. Below its critical feedback strength an axis is unimodal within the noise band $\pm\sqrt{\sigma_a^2/2\gamma}$; above it (dashed) it splits into the complementary leader/follower role modes of the anti-coordination game, which separate further with β . Arrows on the top spine mark the cascade, the successive crossings of the critical feedback strengths $\beta_c^{(1)}, \beta_c^{(2)}, \beta_c^{(3)}$ labelled on the axis. Dotted curves are the linear mean-field theory, which diverges at $\Lambda_k = 1$ (Eq. (6)), while the quartic confinement saturates the actual curve. The right strip shows the status distribution at $\beta = 20$, axis 1 fully bimodal, axis 3 only marginally.

roles differentiate and agents seek to maximize their pay-offs ([26, Sec. S1C]). The maximal coordination gain $w_0 = 4 - 2.5 = 1.5$ per encounter is the analogous work available to an engine that assigns complementary roles through the measurement of social signals. The specific numbers are inessential: any anti-coordination game enters the theory only through this single scalar w_0 , which in turn sets merely the scale of wealth. What is essential is the anti-coordination *class*, in which roles are incentive-compatible: each action must be the best reply to its opposite—given Defer, Go must pay more ($6 > 3$), and given Go, Defer must pay more ($2 > -1$). Even equal off-diagonal payoffs would suffice. We set them unequal to keep the general case. So what kind of engine, built from a multi-agent system, can assign complementary roles?

Agents, game contexts, and identities. N_a agents play the game against each other, selecting actions $a \in$

$\{-1(\text{Defer}), +1(\text{Go})\}$ in pairwise encounters across a repertoire of M distinct game contexts. We emphasize the importance of variable context in capturing both the transposability of schemas and the context-dependent nature of roles. We represent identities, $\boldsymbol{\omega}_i$ for agent i , and contexts, $\mathbf{g}_m$ for context m , as vectors $\boldsymbol{\omega}_i, \mathbf{g}_m \in \mathbb{R}^d$ with embedding dimension d . Each agent stores and presents its own identity in each encounter. Other implementations of the identity state leave the dynamics unchanged: a public registry (Sec. IIA), or platform-estimated identities (Appendix C). For simplicity, the identity of agent i in context m is contextualized as the scalar $\boldsymbol{\omega}_i \cdot \mathbf{g}_m$. When paired with j in an encounter, agent i makes a noisy observation of the *status difference* $\Delta_{im}(j) = (\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m$, namely $s_{im} = \Delta_{im}(j) + \sigma_{\text{read}} \xi$ with ξ a standard Gaussian noise source. Averaged over the drawn partner j , this reduces to Δ_{im} , the form used

below. We call this collective sensing process the *status channel*. In each encounter it communicates a role prescription, the sign of $\Delta_{im}(j)$. The *strategy* of the correlated equilibrium selects Go ($a = +1$; payoff 6) where $\Delta_{im}(j)$ is positive and Defer ($a = -1$; payoff 2) where negative. The deterministic policy $a = \text{sign}(\frac{1}{T} \sum_t s_t)$ makes the Bayes-optimal prescription-following decision under the uncertainty remaining after integrating T samples of evidence, with error rate depending on the magnitude of $\Delta_{im}(j)$ relative to σ_{read} . The term *status* reflects the reward asymmetry of the actions and the resulting identities. That asymmetry is immaterial for the theory, so the term should be seen generally as a mere label for the signal.

Roles require persistent identities. We take a simple choice: each identity low-pass filters its agent's action history, here a context-dependent signed action input. This evolves under a restoring force and identity noise,

$$\dot{\omega}_i = -\gamma \omega_i (1 + |\omega_i|^2) + b_0 \sum_{e \in E_i} a_e \mathbf{g}_{m_e} \delta(t - t_e) + \sigma_\omega \boldsymbol{\eta}_i, \quad (1)$$

where E_i is the set of encounters involving agent i , a_e and m_e the decision and context of encounter e , b_0 the per-encounter kick, γ the decay rate of identity, and $\boldsymbol{\eta}_i$ a vector of standard Gaussian noise sources. This noise represents influences on the identity outside the modeled encounters, and supplies the individual variation the loop amplifies into roles. Equation (1) is an overdamped Langevin equation in a quartic confining potential, driven by encounter events: the restoring force is $-\nabla_{\omega_i} \gamma (|\omega_i|^2/2 + |\omega_i|^4/4)$, whose cubic term serves only to keep $|\omega_i|$ finite (Appendix A). Its coefficient is tied to γ as a choice of units for $|\omega_i|$, absorbed by rescaling $\mathbf{g}_m$ and β , so the linear decay rate is the only free parameter of the restoring force. Encounters are Poisson at rate ν per agent, a fixed interaction budget rather than one that grows with N_a , drawn across contexts with normalized frequencies $\sum_m f_m = 1$ and over partners with pairing probabilities $p_{ij}^{(m)}$ normalized over the partner, $\sum_j p_{ij}^{(m)} = 1$. Averaging each decision over observation noise gives the expected signed action $\langle a \rangle_m$. On timescales long compared with the encounter interval $1/\nu$, and where a single encounter moves an identity by much less than σ_ω^2/β (Appendix A), the kick stream is well approximated by its mean rate over contexts and partners: the dense-encounter limit ($\nu \rightarrow \infty$, $b_0 \rightarrow 0$ at fixed $\beta \equiv b_0 \nu$). In the mean-field limit ($N_a \rightarrow \infty$), and approximating the policy by its value at the partner-averaged status difference (exact as $\sigma_\omega \rightarrow 0$, Sec. II A), the drift becomes

$$\dot{\omega}_i = -\gamma \omega_i (1 + |\omega_i|^2) + \beta \sum_m f_m \langle a \rangle_m \mathbf{g}_m + \sigma_\omega \boldsymbol{\eta}_i, \quad (2)$$

where $\langle a \rangle_m$ is the expected signed action in context m —a sigmoid of the deterministic status difference, Δ_{im} . The agent acts on the observed signal, $a = \pi(s_{im})$. The observation noise is what makes the action random given

Δ_{im} . Averaging the policy over that noise gives the Go-probability $\bar{\pi}(\Delta_{im}) = \mathbb{E}_\xi[\pi(\Delta_{im} + \sigma_{\text{read}}\xi)] = \Pr(a=+1 | \Delta_{im})$, so $\langle a \rangle_m = 2\bar{\pi}(\Delta_{im}) - 1$. Its slope at zero difference defines the *read gain* $g \equiv 2\bar{\pi}'(0)$, which sets how strongly actions are written into memory: thresholding the mean of T Gaussian samples gives $g = \sqrt{2T/\pi}/\sigma_{\text{read}}$. The rate β sets how strongly the social signal drives the identity, and is referred to below as the feedback strength. Pairing is annealed: partners are redrawn at each encounter rather than fixed in a persistent network. The matching rule is otherwise unrestricted, and enters the linearized dynamics through one number, the correlation $\rho_{\text{pair}} \in [-1, 1]$ between a partner's status and one's own at fixed context m , induced by a matching scheme (Sec. II A). Uniform matching sets $\rho_{\text{pair}} = 0$, leaving the partner projection at its population mean, which vanishes in the symmetric state, the undifferentiated population before any role structure exists, so $\langle a \rangle_m$ depends on ω_i alone. Rank-ordered (status-mirrored) matching drives $\rho_{\text{pair}} \rightarrow -1$ and assortative (homophilic) matching drives it to 1, the two ends of the range ([26, Sec. S2 B]).

A persistent network instead enters through $p_{ij}^{(m)}$ as a graph Laplacian (see Appendix A for extensions).

Resources. Coordination income accumulates as agent wealth,

$$\dot{r}_i = \sum_m f_m W_{im} - c r_i, \quad \text{with } W_{im} = w_0 (1 - 2P_{e,im})^2 \quad (3)$$

the encounter-averaged payoff above Nash in context m ([26, Eq. (S2)]), at role-inference error rate $P_{e,im} = \Phi(-|\Delta_{im}|/\sigma_{\text{read}})$ with Φ the standard normal distribution function, and c the resource-decay rate, so $1/c$ is the horizon over which past income still counts. This payoff obeys a Szilard bound, $W \leq 2 \ln 2 w_0 I$ with $I = 1 - H_{\text{bin}}(P_e)$ the role information supplied by the measurement ([26, Sec. S1]). This is the differentiation counterpart of the information bound on consensus order in Ref. [22]. This symmetric form of I is the equiprobable two-role idealization. Heterogeneous status differences and pairwise error rates enter through W_{im} , the partner average of the pairwise payoff $W_{ijm} = w_0 (1 - 2P_{e,ijm})^2$. The average runs over the same pairing statistics $p_{ij}^{(m)}$ that enter Eq. (1), with W_{ijm} set by the magnitude of the status difference to the partner (extensions treated in [26, Sec. S2]). In any case, I bounds the coordination payoff, in information units. Section VI takes up what a thermodynamic accounting would involve.

Schemas. Agents who profit from taking differentiated roles along a schema reinforce it. A schema is a vector, on whose two parts reinforcement acts separately: its direction $\hat{\mathbf{g}}_m$ fixes what the schema is *about*, while its magnitude fixes how strongly that prescription is *institutionalized*. Writing $\mathbf{g}_m = \sqrt{u_m} \hat{\mathbf{g}}_m$ with $u_m = |\mathbf{g}_m|^2$ splits any such dynamics exactly into a *strength sector* governing u_m —the reinforcement of a schema already in the repertoire—and a *transformation sector* rotating $\hat{\mathbf{g}}_m$, the identity space direction along which anti-correlation

in the paired agents' actions confers coordination benefit.

We develop the strength sector here, which suffices for schema-resource duality: it determines which of an available repertoire become institutionalized, in what order, and which agents are enriched. The schemas available to be played form the repertoire, and the environment fixes its context directions $\{\hat{\mathbf{g}}_m\}$. The environment also fixes where the schemas start: $u_m(0) = u_0 e^{\sigma_u z_m}$, with u_0 the seed scale, σ_u the log-dispersion across contexts, and z_m standard normal. For a deployed system σ_u is set by the platform (Appendix C). Restricting a dynamics to magnitudes along a fixed frame is standard practice—amplitude equations for a fixed set of critical modes [27], replicator and Lotka–Volterra dynamics on a fixed set of types [28], the overlap coordinates of associative memories with quenched patterns [29], and the decoupled mode dynamics of deep linear networks under whitened inputs [30] all take this form. Here the restriction is exact rather than approximate: Eq. (4) is one half of a polar split of the vector dynamics. The other directional half, dynamics of the transformation sector, is analyzed in Appendix D where we show the dynamics and resulting solutions depend strongly on what additional assumptions are made about how schema directions interact, a question we consider separately.

A schema's strength u_m acts at each stage of an encounter with it. It scales the status difference read, $\Delta_{im} \propto \sqrt{u_m}$, which lowers the role-inference error rate $P_{e,im}$ against fixed noise σ_{read} . It also fixes the social gain gu_m and hence the feedback strength at which the status distribution in context m bifurcates into Go and Defer role modes (Sec. II A). Contexts therefore activate one at a time as the feedback strength rises, in the order of their strengths u_m . This sequence of bifurcations, the *cascade* (Sec. III), is the focus of much of the rest of the paper.

The strengths also draw on one finite collective capacity: all M schemas are read through the one status channel. Because the status difference read on axis m scales as $\sqrt{u_m}$, the strength u_m is the power that axis puts on the channel, and $U = \sum_m u_m$ is the total power the population uses for status signaling. A channel of finite power limits that total to U_{cap} . We impose the limit softly, as a pressure $\varphi(U)$ common to every schema, negligible while the channel is slack and diverging as $U \rightarrow U_{\text{cap}}$. Fixing U instead would force the capacity to be exhausted and remove the slack regime in which schemas coexist below it (Sec. IV A). Once the total reaches U_{cap} , raising one strength lowers the rest: institutionalization is then the allocation of that capacity across the available games. In addition and independently, we impose that the channel has finite range: the status differences it can represent are bounded, and Δ_{im} grows as $\sqrt{u_m}$, so any single axis is capped, $u_m \leq u_{\text{cap}}$. Where power capacity U_{cap} is not saturated, schemas may reach their own range ceiling u_{cap} without displacing one another (Sec. IV B).

Each strength then grows in proportion to how much resource-weighted structure lies along its own axis, and

decays at a rate common to every schema:

$$\begin{aligned} \dot{u}_m &= 2u_m(\alpha\lambda_m - \kappa - \varphi(U)), \quad \text{with} \\ \lambda_m &= \hat{\mathbf{g}}_m^\top \mathbf{C}_r \hat{\mathbf{g}}_m, \quad \mathbf{g}_m = \sqrt{u_m} \hat{\mathbf{g}}_m, \quad \mathbf{C}_r = \frac{\sum_i r_i \boldsymbol{\omega}_i \boldsymbol{\omega}_i^\top}{\sum_i r_i} \\ &\text{subject to } u_m \leq u_{\text{cap}} \end{aligned} \tag{4}$$

where $\mathbf{C}_r$ is the resource-weighted population covariance, λ_m the resource-weighted variance along context m , κ the context decay rate, α the context reinforcement rate, and φ the budget pressure of the shared channel. Equivalently $d \ln u_m / dt = 2(\alpha\lambda_m - \kappa - \varphi)$, the factor of two because $u_m = |\mathbf{g}_m|^2$ grows at twice the rate of $\mathbf{g}_m$. The two ingredients of λ_m encode Sewell's duality. It is *co-variance*-driven because a schema can organize only the behavioral variation that exists: only the differentiation already present along axis m can be divided into complementary roles, so a schema is reinforced along the axes of differentiation the roles themselves produced. A schema aligned with no variance prescribes nothing and decays at rate κ . It is *resource*-weighted because a schema is validated by the payoff its adherents accrue rather than by raw differentiation: weighting by wealth r_i makes reinforcement contingent on success—the operational form of Sewell's claim that resources, accumulated by enacting a schema, sustain the schema that produced them. With reinforcement proportional to u_m , Eq. (4) selects among existing schemas rather than creating new ones, at a per-unit rate set by λ_m . Because the decay is common to all schemas, the shares u_m/U obey a replicator equation with fitness $\alpha\lambda_m$ exactly. The finite capacity regulates the total (Sec. IV A). Transposability enters as the coherence of the repertoire, the overlaps $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n$, through which differentiation along one axis raises the resource-weighted variance along another (Sec. IV A).

Equations (1), (3) and (4) together form the minimal social technology needed to close the perception-action-reward loop (Fig. 1a) in a way sufficient for social roles to emerge. The resulting multi-agent system dynamics formalizes Sewell's vision and generates duality as a self-consistent fixed-point equation: schemas shape the role structure, roles generate resources, and resource-weighted structure ($\mathbf{C}_r$) selects schemas. The resource weighting makes survival contingent on success: a schema whose axis differentiates agents who profit by it grows, while one whose adherents gain nothing decays at rate κ however much behavior varies along that axis (Sec. II B). The solution set of these equations of motion depends on the parameters and does not generically produce roles.

II. IGNITION AND THE SELF-CONSISTENT STATE

Here we solve the model in two subsections corresponding to two stages. The state is the identities $\{\boldsymbol{\omega}_i\}$ and resources $\{r_i\}$ over agents together with the strengths $\{u_m\}$ over contexts, evolving under Eqs. (2)–(4). The

ω -dynamics (Eq. (2)) is noise-driven, so its stationary states are distributions and we follow them through their moments. The r -dynamics (Eq. (3)) has no noise, but inherits random variation from the identities through the status gaps. The u -dynamics (Eq. (4)) is deterministic, since its resource-weighted variance λ_m is a population average. The ω -drift is odd, $\langle a \rangle_m$ being odd in Δ_{im} , so the role-free density is symmetric under $\omega \rightarrow -\omega$ and its mean vanishes, $\langle \omega \rangle = 0$, at any values for the strengths. Identity noise spreads the population into a unimodal cloud about the origin (Eq. (13)), whose status gaps leave resources at the role-free mean $\bar{r}_{\text{rf}} = O(\sigma_\omega^2)$ ([26, Sec. S2 B]). Stationarity of the u -dynamics is easy to see from Eq. (4): either $u_m = 0$ or $\alpha\lambda_m = \kappa + \varphi$, the invasion threshold. This section takes the channel slack, where $\varphi \approx 0$ and that threshold is $\alpha\lambda_m = \kappa$, and Sec. IV A takes up the filled channel. The three dynamics run on separated timescales, $\gamma \gg c \gg \kappa$, so the strengths see the resources at the stationary value r^* set by the identities. That reduction leaves the identity block untouched, since the ω -dynamics contains no r_i . The reduced state is therefore $(\omega, \{u_m\})$, the identity coordinate of Eq. (2) together with the strengths, and J below is the Jacobian of the noise-averaged drift at $\omega = 0$. Instability there pushes each agent off the origin. The split into roles is symmetric, so the population mean stays zero and the covariance $\mathbf{C}$ registers the structure. The Jacobian is then

$$J = \begin{pmatrix} -\gamma(\mathbf{I}_{d \times d} - \mathbf{\Lambda}) & 0 \\ 0 & 2[\alpha \text{diag}(\boldsymbol{\lambda}) - \kappa \mathbf{I}_{M \times M}] \end{pmatrix},$$

where $\boldsymbol{\lambda}$ stacks the λ_m of Eq. (4). The strength block is written at the seed scale, where the term in $u_m \partial \lambda_m / \partial u_n$ drops. The two zeros are exact. $\partial \dot{\omega} / \partial u_m = 0$ because the ω -dynamics depends on the strengths only through $\mathbf{\Lambda}\omega$, which vanishes at $\omega = 0$. $\partial \dot{u}_m / \partial \omega = 0$ because $\mathbf{C}_r$ is even in ω , since r^* and $\omega\omega^\top$ are both quadratic in it, so the strengths see the identity sector through the covariance $\mathbf{C}$ alone and never through the population mean. The Jacobian is therefore block diagonal, and its spectrum is that of the identity block together with that of the strength block.

Each block carries one of the two threshold expressions of this paper, where one of its eigenvalues crosses zero: $-\gamma(1 - \Lambda_k)$ at $\Lambda_k = 1$, and $2(\alpha\lambda_m - \kappa)$ at the invasion threshold, where a schema enters (Sec. IV A). Section II A computes the identity block that determines ignition by locating the instability from which roles appear, and gives the cascade of thresholds Eq. (6). That block depends on the schema strengths through $\mathbf{\Lambda}$, evaluated at their instantaneous values, so the criterion holds whether or not the strengths are stationary and we can solve that block for given, fixed strengths. Then, in Section II B, we restore the schemas' own dynamics and solve Eqs. (3) and (4) for their stationary state. We express $\mathbf{C}_r$ in terms of $\mathbf{\Lambda}$ and $\mathbf{C}$, and substitute for $\mathbf{C}$ the subcritical expression Eq. (13) of Sec. II A, valid for $\Lambda_1 < 1$, so the loop closes below the first bifurcation. We state

and summarize these results here before their derivation in subsequent subsections.

The linearization gives the expected social feedback to the identity as $\gamma \mathbf{\Lambda} \omega_i$, with the *social loop gain*

$$\mathbf{\Lambda} \equiv \underbrace{\frac{2\beta}{\gamma}}_{\text{identity persistence}} \cdot \underbrace{\eta_{\text{agent}}}_{\text{cognitive capacity}} \cdot \underbrace{\frac{1 - \rho_{\text{pair}}}{\sigma_{\text{read}}}}_{\text{channel fidelity}} \underbrace{\mathbf{H}^\top \mathbf{H}}_{\text{repertoire}}, \quad (5)$$

where $\mathbf{H}$ stacks the schema vectors $\mathbf{g}_m$, $\eta_{\text{agent}} \equiv \sigma_{\text{read}} g / 2$ is the agent's dimensionless inference efficiency, and $(1 - \rho_{\text{pair}})$ is the pairing gain factor set by the matching scheme's partner correlation of Sec. I. The first three factors are scalars, so they scale every eigenvalue of $\mathbf{\Lambda}$ alike and set the overall gain without distinguishing axes. Only the repertoire factor $\mathbf{H}^\top \mathbf{H}$ separates them: the symmetric state destabilizes along eigenvector k of $\mathbf{\Lambda}$ when the corresponding eigenvalue reaches one,

$$\Lambda_k > 1, \quad \Lambda_1 \geq \Lambda_2 \geq \dots \geq \Lambda_R, \quad (6)$$

with $R = \text{rank}(\mathbf{H}) \leq \min(M, d)$, short of M when the pool outruns the identity dimension or the context directions are linearly dependent. Writing $\mu_1 \geq \dots \geq \mu_R > 0$ for the eigenvalues of $\mathbf{H}^\top \mathbf{H}$, which are the schema strengths u_k themselves when the contexts are orthogonal, Λ_k is the leading gain Λ_1 scaled by μ_k / μ_1 , so the ordered $\{\mu_k\}$ set the cascade and Λ_1 sweeps it. The axes ignite one at a time where the eigenvalues are distinct, and together where the spectrum is degenerate. Here and in Sec. III the strengths are at their initial values, the environment's seeding of Sec. I, held fixed while β sweeps. Section IV makes them dynamical, so the spectrum then rises at fixed β and crosses the same condition $\Lambda\mu_k = 1$ from below. We make the population covariance $\mathbf{C} = N_a^{-1} \sum_i \omega_i \omega_i^\top$ the order parameter. Its eigenvalues v_k leave the noise floor $\sigma_\omega^2 / 2\gamma$ one at a time. The number that have crossed, the height of the cascade N , is the *cultural thickness*: the count of active (supra noise floor) role dimensions.

For a single axis of unit strength $\mathbf{\Lambda}$ is a number and Eq. (6) is the loop-gain criterion of Fig. 1a, $\Lambda > 1$. Ignition names the crossing of that criterion: above it the loop returns more differentiation than identity decay removes, so role structure sustains itself with no external drive. Here η_{agent} is $1/\sqrt{2\pi}$ for a single observation and $\sqrt{T/2\pi}$ with memory T . $\eta_{\text{agent}} \rightarrow 0$ is choice at random and $\eta_{\text{agent}} \rightarrow \infty$ deterministic payoff maximization, the limits the inverse temperature spans in the free-energy account of bounded rationality [31]. The pairing enters the loop gain only through the factor $(1 - \rho_{\text{pair}})$. Random matching is the reference, factor 1. Rank-ordered matching gives factor 2 in the population limit and halves the threshold, and assortative matching gives factor 0 ([26, Sec. S2]). The correlation needs no pre-existing role modes. ρ_{pair} is a property of the matching algorithm applied to whatever status distribution currently exists, including a unimodal and arbitrarily narrow one, so it is

as well defined below threshold, where it sets the thresholds, as above it. Per-context play frequencies or matching reweight $\mathbf{H}^\top \mathbf{H}$ before its eigenvalues are taken, and then the channel factor no longer comes out in front ([26, Sec. S2 C]).

The engine runs on two integrators—a write path (β/γ : decisions accumulated into persistent identity) and a read path (η_{agent} : social signals accumulated into posterior inference)—coupled by the information channel between agents. The factors compensate at linear order: well-designed matching can compensate for limited-inference agents and vice versa (similar to the measurement-quality bound on engine performance derived in [17]). However, even at the best (rank-ordered) matching ($\rho_{\text{pair}} \rightarrow -1$) ignition requires $\eta_{\text{agent}} > \sigma_{\text{read}}\gamma/4\beta$, an *irreducible cognitive floor* below which no interaction orchestration can sustain role structure. At threshold where linear order is accurate, only the product Λ is physical. Beyond this threshold the degeneracy of the factorization breaks: β sets how far the roles separate, independently of ρ_{pair} , while ρ_{pair} sets how often that separation converts to payoff. Role societies at equal Λ but different splits are therefore dynamically distinguishable ([26, Sec. S2]). Figure 1b shows the first three bifurcations in the cascade: equilibrating the population at each β , role axes leave the noise floor one at a time around the critical feedback strengths of Eq. (6).

A. The social loop gain and the cascade of thresholds

We now linearize the mean-field drift Eq. (2) about the symmetric state, showing how Λ arises. Its feedback term is $\beta \sum_m f_m \langle a \rangle_m \mathbf{g}_m$, with $\langle a \rangle_m(\boldsymbol{\omega}_i)$ the expected signed action in context m , averaged over the observation noise and the drawn partner (Sec. I).

Mean field. As $N_a \rightarrow \infty$ the empirical distribution of the other agents converges to a fixed law, so the drawn partner is independent of the focal agent's own history. The matching scheme fixes the partner correlation $\rho_{\text{pair}} = \text{Corr}(x_i, x_j)$ between the projections $x_i = \boldsymbol{\omega}_i \cdot \mathbf{g}_m$ and $x_j = \boldsymbol{\omega}_j \cdot \mathbf{g}_m$, and near the symmetric state the pair is jointly Gaussian, so $\mathbb{E}[x_j | x_i] = \rho_{\text{pair}} x_i$ and the expected status difference is $(1 - \rho_{\text{pair}})x_i$ ([26, Sec. S2 B]). Random matching leaves the unconditional population mean, which vanishes at the symmetric fixed point, and is the case $\rho_{\text{pair}} = 0$. Writing $\Delta_{im} \equiv \boldsymbol{\omega}_i \cdot \mathbf{g}_m$ for the focal projection, the observed signal is $s_{im} \approx (1 - \rho_{\text{pair}})\Delta_{im} + \sigma_{\text{read}}\xi_{im}$, and, averaging the decision over the observation noise,

$$\langle a \rangle_m(\boldsymbol{\omega}_i) = 2\pi((1 - \rho_{\text{pair}})\Delta_{im}) - 1, \quad (7)$$

which for the single-sample policy is $\text{erf}((1 - \rho_{\text{pair}})\Delta_{im}/(\sigma_{\text{read}}\sqrt{2}))$. Substituting the conditional mean inside the policy is exact as $\sigma_\omega \rightarrow 0$, where the cloud shrinks to a point at the origin and the threshold

is set; the correction away from onset is $O(\sigma_\omega^2)$ ([26, Sec. S6]).

Linearization. At $\boldsymbol{\omega}_i = \mathbf{0}$ each projection vanishes and $2\pi(0) - 1 = 0$ by policy symmetry. To first order $\langle a \rangle_m \approx g(1 - \rho_{\text{pair}})(\mathbf{g}_m^\top \boldsymbol{\omega}_i)$, so the feedback term of Eq. (2) becomes

$$\begin{aligned} \beta \sum_m f_m \langle a \rangle_m \mathbf{g}_m &\approx \beta g(1 - \rho_{\text{pair}}) \left(\sum_m f_m \mathbf{g}_m \mathbf{g}_m^\top \right) \boldsymbol{\omega}_i \\ &= \gamma \Lambda \boldsymbol{\omega}_i, \end{aligned} \quad (8)$$

$$\begin{aligned} \Lambda &\equiv \Lambda \sum_m f_m \mathbf{g}_m \mathbf{g}_m^\top = \Lambda \mathbf{H}^\top \text{diag}(\mathbf{f}) \mathbf{H}, \\ \Lambda &\equiv \beta(1 - \rho_{\text{pair}})g/\gamma, \end{aligned} \quad (9)$$

with $\mathbf{f} = (f_1, \dots, f_M)$, $\mathbf{H}$ the $M \times d$ matrix of stacked context vectors, and Λ the scalar loop gain of a unit-strength axis. We take the play frequencies uniform, $f_m = 1/M$, and absorb the constant $1/M$ into β , so that Λ takes the form of Eq. (5) and β is the per-context write rate. The frequencies are retained explicitly only where they act as a design lever ([26, Sec. S2 D]). They are properties of the environment, fixed over the full set of M candidate contexts, and are not renormalized when a schema's strength decays away. Its context is still played, so extinction does not concentrate weight on the survivors. Combined with the linearized restoring force (the cubic term vanishes at the origin),

$$\dot{\boldsymbol{\omega}}_i \approx -\gamma(\mathbf{I}_{d \times d} - \Lambda)\boldsymbol{\omega}_i + \sigma_\omega \boldsymbol{\eta}_i. \quad (10)$$

The matching scheme enters only through the factor $(1 - \rho_{\text{pair}})$, which multiplies every axis alike and vanishes at $\rho_{\text{pair}} = 1$. The symmetric state thus loses stability along eigenvector k of Λ when its eigenvalue reaches one (Eq. (6)), so the thresholds are

$$\Lambda_k = \Lambda_1 \frac{\mu_k}{\mu_1} = 1, \quad \mu_1 \geq \mu_2 \geq \dots \geq \mu_d, \quad (11)$$

the cascade of thresholds, crossed in the order of the repertoire's eigenvalues.

The subcritical state. At fixed schemas the drift is the gradient $-\nabla_\omega E$, with

$$\begin{aligned} E(\boldsymbol{\omega}) &= \gamma \left(\frac{|\boldsymbol{\omega}|^2}{2} + \frac{|\boldsymbol{\omega}|^4}{4} \right) - \beta \sum_m \Psi(\boldsymbol{\omega} \cdot \mathbf{g}_m), \\ \Psi'(z) &= 2\pi((1 - \rho_{\text{pair}})z) - 1, \end{aligned} \quad (12)$$

so the stationary distribution is Boltzmann, $p^*(\boldsymbol{\omega}) \propto \exp(-2E(\boldsymbol{\omega})/\sigma_\omega^2)$, and identities are i.i.d. draws. Dropping the quartic term recovers the linear drift of Eq. (10), so near threshold p^* is Gaussian with covariance (Lyapunov equation, symmetric drift)

$$\mathbf{C} = \frac{\sigma_\omega^2}{2\gamma}(\mathbf{I}_{d \times d} - \Lambda)^{-1}, \quad (13)$$

valid for $\Lambda_1 < 1$. $\mathbf{C}$ diverges along the leading eigenvector of $\mathbf{\Lambda}$ at the bifurcation. Each covariance eigenvalue $v_k = \sigma_\omega^2 / 2\gamma(1 - \Lambda_k)$ increases with the loop gain Λ_k it comes from, so Eq. (13) can be inverted: measuring the covariance spectrum below threshold recovers $\{\Lambda_k\}$, and with it the cascade still to come (Sec. V).

Validity of the linearization. Stability is a Jacobian property evaluated at the fixed point, independent of where the probability mass sits, so Eq. (11) is the exact stability boundary at any noise level. The cloud's width enters only through the partner average, which softens the gain and shifts onset by $O(\sigma_\omega^2)$. This is a mean-field Ginzburg correction that vanishes with σ_ω and is negligible outside a shrinking critical window ([26, Sec. S6]).

B. The self-consistent fixed point

Section II A solved the identity sector. The two slower dynamics close the loop. Equations (2)–(4) evolve on separated timescales $\gamma \gg c \gg \kappa$, so identities equilibrate first, then resources, then schemas, and each is stationary with respect to the ones below it.

Step 1: resources. At steady state $r_i^* = c^{-1} \sum_m f_m \mathbb{E}_j[W_{ijm}]$, a deterministic function of ω_i . Near threshold the error rate linearizes, $P_e \approx \frac{1}{2} - |(\omega - \omega') \cdot \mathbf{g}_m| \bar{\pi}'(0)$, so $W_{ijm} \approx w_0 g^2 ((\omega_i - \omega_j) \cdot \mathbf{g}_m)^2$. Averaging over the drawn partner, jointly Gaussian with the focal projection at correlation ρ_{pair} ($\mathbb{E}[x_j | x_i] = \rho_{\text{pair}} x_i$ and $\text{Var}[x_j | x_i] = (1 - \rho_{\text{pair}}^2) \mathbf{g}_m^\top \mathbf{C} \mathbf{g}_m$, with $x_i = \omega_i \cdot \mathbf{g}_m$),

$$r^*(\omega) \approx \frac{w_0 g^2}{c} \sum_m f_m \left((1 - \rho_{\text{pair}})^2 (\omega \cdot \mathbf{g}_m)^2 + (1 - \rho_{\text{pair}}^2) \mathbf{g}_m^\top \mathbf{C} \mathbf{g}_m \right). \quad (14)$$

The first term is agent-specific; the second a population baseline. Random matching removes both factors.

Step 2: resource-weighted covariance. With $\mathbf{C}_r = \int r^*(\omega) \omega \omega^\top p^* / \int r^*(\omega) p^*$ and r^* quadratic in ω , the numerator is a fourth moment, evaluated for Gaussian p^* by Wick's theorem (Isserlis' theorem in the statistics literature),

$$\mathbb{E}[(\omega \cdot \mathbf{g}_m)^2 \omega_a \omega_b] = (\mathbf{g}_m^\top \mathbf{C} \mathbf{g}_m) C_{ab} + 2(\mathbf{g}_m^\top \mathbf{C})_a (\mathbf{g}_m^\top \mathbf{C})_b, \quad (15)$$

making $\mathbf{C}_r$ a function of $\mathbf{C}$ and $\{\mathbf{g}_m\}$ alone. Even when $\mathbf{C} \propto \mathbf{I}_{d \times d}$, $\mathbf{C}_r$ is anisotropic, because $r^*(\omega) \propto \sum_m f_m (\omega \cdot \mathbf{g}_m)^2$ weights agents whose identities align with the context directions more heavily. This anisotropy resolves the collapse of the unweighted dynamics ($r_i \equiv 1$, $\mathbf{C}_r = \mathbf{C}$): $\mathbf{C}_r$ inherits structure from the repertoire even when the population is isotropic. Summing Eq. (15) over contexts gives a closed form for the resource-weighted covariance, valid for any repertoire geometry and unchanged when $\mathbf{\Lambda}$ is rescaled,

$$\mathbf{C}_r = \mathbf{C} + (1 - \rho_{\text{pair}}) \frac{\mathbf{C} \mathbf{\Lambda} \mathbf{C}}{\text{tr}(\mathbf{C} \mathbf{\Lambda})}. \quad (16)$$

Since $\mathbf{C} = \frac{\sigma_\omega^2}{2\gamma} (\mathbf{I}_{d \times d} - \mathbf{\Lambda})^{-1}$ commutes with $\mathbf{H}^\top \mathbf{H}$, $\mathbf{C}_r$ has the same eigenbasis as $\mathbf{H}^\top \mathbf{H}$. The two prefactors of Eq. (14) combine into the single factor $(1 - \rho_{\text{pair}})$ ([26, Sec. S2 B]), so assortative matching ($\rho_{\text{pair}} \rightarrow 1$) leaves $\mathbf{C}_r = \mathbf{C}$ and removes the resource weighting altogether, while rank-ordered matching doubles it. For orthogonal contexts Eq. (16) collapses to a scalar relation on each axis,

$$\lambda_m = v_m \left(1 + (1 - \rho_{\text{pair}}) \frac{u_m v_m}{\sum_n u_n v_n} \right), \quad (17)$$

with $v_m = \frac{\sigma_\omega^2}{2\gamma(1 - \Lambda_m)},$

where the strength enters through $\Lambda_m = \Lambda u_m$. The resource weighting fixes that denominator: normalizing $\mathbf{C}_r$ by $\sum_i r_i$ puts the population's mean income there, so $\sum_n u_n v_n$ is the role-free mean $\bar{r}_{\text{rf}}$ up to constants ([26, Sec. S2 B]).

Step 3: schemas. At steady state each surviving schema satisfies $\alpha \lambda_m = \kappa$, with $\lambda_m = \hat{\mathbf{g}}_m^\top \mathbf{C}_r \hat{\mathbf{g}}_m$ the resource-weighted variance along its (fixed) axis. Those with $\lambda_m < \kappa/\alpha$ decay to zero. Steps 1–3 therefore close into a condition on the repertoire alone,

$$\alpha \lambda_m(\{\mathbf{g}_n\}) = \kappa \quad \text{for every surviving } m, \quad (18)$$

since Eqs. (13) and (16) fix $\mathbf{C}_r$ from $\{\mathbf{g}_n\}$ and nothing else. Schemas set the identities whose resource-weighted covariance selects the schemas, which is Sewell's duality as a fixed-point equation. For orthogonal contexts Eq. (17) makes it explicit in the strengths $\{u_n\}$.

The loop and multiplicity. The conditions close: $\{\mathbf{g}_m\} \rightarrow E \rightarrow p^* \rightarrow \mathbf{C} \rightarrow r^* \rightarrow \mathbf{C}_r \rightarrow \{\mathbf{g}_m\}$. The trivial solution $\mathbf{g}_m^* = 0$ always exists. Nontrivial solutions appear when an eigenvalue of $\mathbf{C}_r$ exceeds κ/α . Because $\mathbf{C}_r$ is anisotropic even at isotropic $\mathbf{C}$, those eigenvalues differ from the outset, so the condition selects among schemas rather than admitting or rejecting all of them. Since $\mathbf{C}_r$ has d eigenvalues, the sustained schema count is

$$M_{\text{eff}} = \#\{k : \lambda_k(\mathbf{C}_r) \geq \kappa/\alpha\} \leq d, \quad (19)$$

the number of schemas viable at this fixed point, those whose resource-weighted variance clears the decay rate. It caps the cascade: N cannot exceed the number of axes available to activate, and reaches M_{eff} as β grows. How many of them the budget can sustain, and whether the endpoint is a saturated cluster or a single condensate, is decided by the capacity ratio of Sec. IV A. Without resources ($r_i \equiv 1$, $\mathbf{C}_r = \mathbf{C}$) the λ_m rank the fixed axes by their alignment with the leading eigenvector of $\mathbf{C}$, so the survivor set narrows toward the best-aligned schema. The resource reweighting, anisotropic along the context directions themselves, lifts several λ_m over κ/α at once and sustains a multi-schema state. This fixed point identifies the surviving schemas but leaves the shape of their spectrum implicit. Since this shape determines the cascade, Section III asks what an arbitrary spectrum implies for the cascade, and classifies the possible cascades.

Section IV analyzes the capacity-constrained dynamics and determines which spectra and cascade result. The fixed point organizes that analysis rather than describing a resting state: Sec. IV A shows it is a saddle, and the dynamics leave it along the unstable direction.

III. THE FUNCTIONAL FORM OF THE CASCADE

Role-based societies emerge as the loop gain rises with time, $\Lambda(t)$, summarizing the culture and technology innovations that improve identity persistence, inference, or channel fidelity. Only the product is physical at threshold, so raising any factor of Λ serves equally. Our simulations sweep β , which enters Λ linearly and enters nothing else. We take the sweep to be slow compared with identity relaxation, $\dot{\Lambda}/\Lambda \ll \gamma$, so the population tracks it. What follows are the cascades possible for any such fixed spectrum. Section IV asks which spectra the loop produces.

The number of active axes at feedback strength β is the counting function $N(\beta)$. Here, we derive the effective local rate of growth of this function. Both follow from the shape of the eigenvalue spectrum ρ of the repertoire $\mathbf{H}^\top \mathbf{H}$ in the limit $M, d \rightarrow \infty$ at fixed aspect ratio M/d . We find that only three repertoire properties shape $N(\beta)$: the strength hierarchy $\{u_m\}$, the coherence $\{\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n\}$, and the aspect ratio M/d . Equation (4) makes the first endogenous and leaves the other two to the environment. These generate five classes of spectrum type (with corresponding cascade shape) (Fig. 2d). Figure 2a,c give an example of each at $R = 50$ modes, while Fig. 2b shows the rate for $R \rightarrow \infty$.

A. The counting function and acceleration criterion

Let $\mu_1 \geq \dots \geq \mu_R > 0$ be the nonzero eigenvalues of $\mathbf{H}^\top \mathbf{H}$, $R = \text{rank}(\mathbf{H}) \leq \min(M, d)$, so that $\Lambda_k = \Lambda \mu_k$ with Λ the scalar prefactor of Eq. (5). Axis k bifurcates at $\Lambda = 1/\mu_k$ (Eq. (6)). The number of active axes at loop gain Λ is

$$N(\Lambda) = \#\{k : \mu_k > 1/\Lambda\} = R \bar{F}(1/\Lambda), \quad (20)$$

$$\bar{F}(\mu) = \int_\mu^\infty \rho(\mu') d\mu', \quad (21)$$

with ρ the spectral density of $\mathbf{H}^\top \mathbf{H}$: the cascade is the pushforward of the spectrum under $\mu \mapsto 1/\Lambda$. The rate is

$$\frac{dN}{d\Lambda} = \frac{R}{\Lambda^2} \rho\left(\frac{1}{\Lambda}\right), \quad \frac{dN}{d \ln \Lambda} = R \mu \rho(\mu) \Big|_{\mu=1/\Lambda}. \quad (22)$$

Defining the local spectral exponent $\zeta(\mu) \equiv -d \ln \rho / d \ln \mu$, differentiating again gives the accel-

eration criterion

$$\frac{d^2 N}{d\Lambda^2} > 0 \iff \frac{d}{d\mu} [\mu^2 \rho(\mu)] < 0 \iff \zeta > 2 \text{ at } \mu = 1/\Lambda. \quad (23)$$

Early in a sweep only the largest μ_k have ignited, so a slow start says nothing about ζ further down the spectrum. Reading it as a slow cascade is an assumption about the unseen modes there, a central difficulty for monitoring addressed in Sec. V.

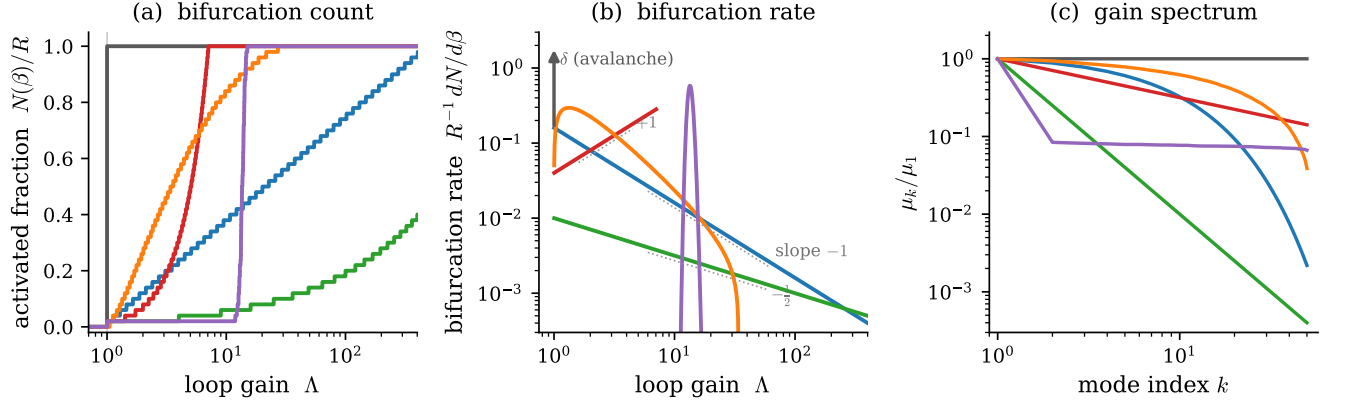
B. The five exogenous classes

Figure 2d catalogs the qualitatively distinct spectral classes and the cascade each induces. Any repertoire's spectrum is locally one of these, and exactly one where the spectrum is quasistatic across the sweep. Only three repertoire properties shape ρ : the strength hierarchy $\{u_m\}$, the coherence $\{\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n\}$, and the aspect ratio $q = M/d$. Each class is the canonical form taken when one of the three properties dominates. The strength hierarchy $\{u_m\}$ alone sets μ_k when contexts are effectively orthogonal: flat, exponential, and power-law forms give classes (i)–(iii), and one strength dominating a flat bulk gives class (v). The aspect ratio $q = M/d$ alone shapes class (iv), whose Marčenko–Pastur edges are functions of q . The coherence $\{\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n\}$, through its uniform value χ , reaches class (v) by the other route, redundancy rather than dominance.

The boundary $a = 1$ ($\zeta = 2$), with a the Zipf exponent of the strength hierarchy, separates the decelerating and accelerating a Zipf-shaped spectrum. The random class [32] has its spectrum confined to $[\mu_-, \mu_+]$, with the edges set by the aspect ratio $q = M/d$. Bifurcations run from the largest eigenvalue down, so the cascade enters the bulk at its upper edge, where the density vanishes as $\rho \sim \sqrt{\mu_+ - \mu}$ —hence the vanishing onset rate $dN/d\Lambda \propto (\Lambda - 1)^{1/2}$ before the bulk. The spiked class comes from redundancy, coherent contexts sharing directions. At uniform coherence χ and common strength $u = |\mathbf{g}_m|^2$, $\mathbf{H}^\top \mathbf{H}$ has one outlier $u(1 + (M - 1)\chi)$ over a degenerate bulk $u(1 - \chi)$, and the gain scales both by g . The quiescent interval therefore has relative width $(1 + (M - 1)\chi)/(1 - \chi)$, and an avalanche follows. Rate data alone cannot distinguish this interval from a cascade that has ended.

IV. ENDOGENOUS SELECTION OF THE REPERTOIRE

In Sec. III the schema strengths, and thus the cascade type, were taken as given. $\{u_m\}$ are set instead by the slower schema-resource dynamics of Eqs. (3) and (4), both at the stationary state and along the transient that approaches it. The context directions $\{\hat{\mathbf{g}}_m\}$ stay exogenous throughout (Sec. I). Here we classify that solution space by the resulting cascade type, and map which



Cascade class	$N(\beta)$	Spectrum μ_k	Set by	For a monitor
— (i) single avalanche	step	degenerate $\mu_k = \mu$	strengths flat	no warning
— (ii) logarithmic	$k_0 \ln \Lambda$	geometric $\mu_k \propto e^{-k/k_0}$	strengths exponential	steady per e -fold
— (iii) decelerating	$\Lambda^{1/a}$	Zipf $a > 1$ $\mu_k \propto k^{-a}$	strengths power-law	informative early
— (iii) accelerating	$\Lambda^{1/a}$	Zipf $a < 1$ $\mu_k \propto k^{-a}$	strengths power-law	early rate too low
— (iv) quiet, then burst	$(\Lambda - 1)^{3/2}$	random	Marčenko–Pastur aspect ratio q	loop still open
— (v) one, gap, then all	gap, then jump	spiked outliers + bulk	strengths spiked, or coherence χ	quiet mimics an end

FIG. 2. **Cascade classes of exogenous schema repertoires.** (a) The number of active role axes $N(\beta)$. (b) The bifurcation rate, which accelerates where its slope is positive, equivalently where $\zeta > 2$ (Eq. (23)). (c) The rank-ordered gain spectrum, normalized to its leading eigenvalue; (a) is its pushforward under $\mu \mapsto 1/\Lambda$. (d) The classes and their cascades, the last column anticipating the monitoring analysis of Sec. V. All four are given over the five spectral classes of the repertoire $\mathbf{H}^T \mathbf{H}$, keyed by the colours of (d): (i) degenerate spectra give a single avalanche with no advance warning; (ii) geometric hierarchies give a logarithmic cascade with constant bifurcations in log-space; (iii) power-law (Zipf) hierarchies split at exponent $a = 1$ into decelerating (informative early phase) and accelerating (early rate under-predicts) cascades, with log-log slope $(1 - a)/a$ in panel (b); (iv) unstructured random repertoires (Marčenko–Pastur) start quietly, $dN/d\Lambda \propto (\Lambda - 1)^{1/2}$, then burst; (v) coherent (spiked) repertoires produce an early bifurcation, a quiescent interval, then an avalanche. The curves use $k_0 = 8$, $a = 2$ and $a = \frac{1}{2}$, $q = 2$, and $\chi = 0.2$, all drawn with the same $R = 50$ modes, the random class as 50 quantiles of its bulk, so only the spectral shape differs. The gap below the leading mode in (c) is what the quiescent interval reads out, and the crowding of the random spectrum against its upper edge is what makes the early rate vanish.

regimes of schema-resource dynamics exhibit which cascade types across the transition to a role-based society.

Two ratios separate the regimes, one from the transient and one from the stationary state. Along the transient, $r = r_{\text{act}}/r_{\text{comp}}$ counts the axes that cross threshold per compounding time $1/r_{\text{comp}}$, and is fixed by the dispersion of the seeded repertoire (Sec. IV B). At the stationary state, $\Theta = M/M_{\text{cap}}$ counts the schemas competing per capacity slot, with $M_{\text{cap}} = U_{\text{cap}}/u_{\text{cap}}$ the number of schemas that the capacity sustains at the per-schema ceiling. Large Θ divides the budget more finely, so less of it reaches any one schema. Below $\Theta = 1$ the per-schema ceiling binds first and all M schemas survive, the *thick band*; above it the shared budget runs out and only M_{cap} do.

A. The closed-pool (survivor) construction

Under the full loop the strength dynamics produce the spectrum. We treat the M context directions as can-

didate schemas and ask how many survive. The pool is *closed*: no schema is added after seeding, so the dynamics can only winnow it. Schema innovation is an open-repertoire process we do not treat (Sec. IV B).

The active set. The active set is $\{m : u_m > u_0\}$ with $u_m = |\mathbf{g}_m|^2$ and u_0 the seed scale, the small strength at which schemas are seeded and above which they count as active. Its size $K \leq M$ is the emergent survivor count, read at the end of the sort (Sec. IV B). Orthogonal contexts make each $\hat{\mathbf{g}}_m$ an eigenvector of $\mathbf{C}_r$ (Sec. II B), so λ_m is an eigenvalue of $\mathbf{C}_r$. Fixed axes keep distinct survivors on distinct eigen-directions: a schema cannot rotate onto another, so two survivors share a direction only if the repertoire supplies them one. In this case, K equals M_{eff} (Eq. (19)), the number of eigenvalues of $\mathbf{C}_r$ that clear κ/α . Under the vector dynamics of Appendix D K and M_{eff} would instead separate, since every schema rotates toward the same leading eigenvector. The seeding of Sec. I is centered on u_0 , so half the repertoire starts above it at any σ_u . Only the sort separates the u_m into a saturated group and a group decaying to zero, and the

count is insensitive to u_0 once it has. But u_0 does enter the regime boundaries through the log dynamic range $L = \ln(u_{\text{cap}}/u_0)$ of Sec. IV B.

The strength replicator. Written in log form, the strengths of Eq. (4) obey

$$\frac{d}{dt} \ln u_m = 2(\alpha\lambda_m(\mathbf{C}_r) - \kappa - \varphi(U)), \quad U = \sum_n u_n, \quad (24)$$

a replicator dynamics on the shares with per-schema fitness $\alpha\lambda_m$ [33, 34]—the total reinforcement of schema m being $\mathbf{g}_m^T \mathbf{C}_r \mathbf{g}_m = u_m \lambda_m$ —and the budget pressure $\varphi(U)$ of Sec. I, with $\varphi \approx 0$ while the channel is slack and $\varphi' > 0$ as it fills.

Why the budget is soft. The bare κ -only dynamics ($\varphi \equiv 0$) leaves U unregulated—since λ_m grows with u_m (below), a schema past the invasion threshold would grow indefinitely. φ supplies the missing restoring force, equally on every axis, so it shifts the invasion threshold without biasing which schemas survive. The budget is a carrying capacity, not a conservation law: attention is capped rather than conserved and redistributed, and it can be left under-used. With κ a fixed decay rate and the budget a soft inequality ($U \leq U_{\text{cap}}$), two regimes survive that the normalized replicator would eliminate. The first is a sub-threshold state in which every schema decays, the pre-role condition the ignition onset transitions out of. The second is a slack regime ($\Theta < 1$, $U < U_{\text{cap}}$) in which all M schemas coexist at the per-schema ceiling, the durable multiplicity central to the Sewellian picture. Relative selection admits interior coexistence. A single winner comes instead from the increase of λ_m with u_m (below).

The invasion threshold. A schema enters the active set where its replicator rate Eq. (24) turns positive, $\alpha\lambda_m > \kappa + \varphi$. The correction term of Eq. (17) is nonnegative for $\rho_{\text{pair}} \leq 1$, so $\lambda_m \geq v_m$, and both grow with u_m . The invasion strength $u_m^\dagger$ therefore satisfies

$$\Lambda u_m^\dagger \leq 1 - \frac{\alpha\sigma_\omega^2}{2\gamma(\kappa + \varphi)} < 1,$$

with equality as the invading schema's share of $\sum_n u_n v_n$ vanishes. Since λ_m increases with u_m , $u_m^\dagger$ is unstable: a schema seeded below it decays to zero, and one seeded above it grows. Growth continues until the strength reaches $1/\Lambda$ and the axis bifurcates, or until a capacity ceiling arrests it short of that (Sec. IV B). Budget pressure raises $u_m^\dagger$, so schemas already drawing on the channel lift the threshold a later entrant must clear.

Invasion precedes ignition, since $u_m^\dagger$ sits below the bifurcation strength $1/\Lambda$ of Eq. (6) by the noise floor $\sigma_\omega^2/2\gamma$ measured against the decay rate, and the two coincide only as $\sigma_\omega \rightarrow 0$. The strengths therefore arrive at the ignition threshold under their own dynamics rather than resting at it. That arrival is the sort of Sec. IV B, and it is why the thresholds of Sec. II A apply at instantaneous strengths rather than only at stationary ones. Below every bifurcation the covariance sits at the noise floor of

Eq. (13) rather than at zero, so $\lambda_m > 0$ and the strengths grow before any role structure exists. The same identity noise sets the separation between the two thresholds. A positive $u_m^\dagger$ requires $\alpha\sigma_\omega^2/2\gamma < \kappa + \varphi$, and where that fails every seeded schema invades, so the sub-threshold state in which every schema decays holds only while the decay rate exceeds the fluctuation reinforcement.

The pinning saddle. With Eq. (17) of Sec. II B in hand, two competing results follow.

1. *The pinning condition.* A schema is stationary only where its replicator rate Eq. (24) vanishes, $\alpha\lambda_m = \kappa + \varphi^*$ with $\varphi^* = \varphi(U^*)$ the equilibrium budget pressure, so at an interior fixed point all survivors share one $\lambda_m = (\kappa + \varphi^*)/\alpha$. The shift φ^* is common to every axis, so equal fitness still demands equal strength. λ_m is strictly increasing in u_m : both factors in Eq. (17) grow with u_m across the sub-critical range $\Lambda_m < 1$, a property that extends to non-orthogonal repertoires ([26, Sec. S5]). The interior fixed point therefore has a degenerate spectrum. This is the competitive-exclusion result for replicators. Monotonicity sharpens it from equal fitness to equal strength.
2. *The condensation instability.* The pinned point is a saddle, and both of its directions follow from Eq. (24) directly. Along the *magnitude* direction $u_m = u + \delta$ (all schemas together), growth is arrested by whichever capacity ceiling is reached first (Sec. IV B): the per-schema cap u_{cap} , or the total budget through $\varphi(U)$, which subtracts a restoring $2M\varphi'(U^*)\delta$ from every growth rate against the destabilizing $2\alpha(\partial\lambda/\partial u)\delta$. This stabilizes the direction once $M\varphi'(U^*) > \alpha\partial\lambda/\partial u$. Either way an imposed ceiling supplies the restoring force that fixes U ; the bare dynamics do not. Along a *composition* direction $\sum_m \delta u_m = 0$ the total, and hence φ , are unchanged, and Eq. (24) linearizes to $\delta\dot{u}_m \propto \delta\lambda_m \propto \delta u_m - \bar{\delta}u$: because λ_m increases with u_m , any schema pushed above the mean grows, draining the rest—*unstable*. One stable magnitude direction and $M - 1$ unstable composition directions make the equal-strength state a saddle (Fig. 3), and the dynamics leave it toward condensation. The departure is autocatalytic near a bifurcation, where v_m (and hence $\partial\lambda_m/\partial u_m$) diverges. There the budget pressure arrests the runaway at a saturated cluster, and so regulates condensation. Whether it does so (a cluster) or does not (a bare condensate) is set by $\Theta = Mu_{\text{cap}}/U_{\text{cap}}$, mapped globally in Fig. 4.

Classification of stationary states. At a fixed point of Eq. (24) every schema is extinct ($u_m = 0$, $\alpha\lambda_m < \kappa + \varphi^*$), marginal ($0 < u_m < u_{\text{cap}}$, $\alpha\lambda_m = \kappa + \varphi^*$, at the invasion threshold), or saturated ($u_m = u_{\text{cap}}$). Since pinning forces equal fitness to equal strength, no stationary state has a graded active distribution, so there are exactly two fixed-point families. (a) The *degenerate* state: all survivors equal, stabilized by the ceiling into a cluster of $\min(M, M_{\text{cap}})$ saturated schemas. Its flat spectrum bi-

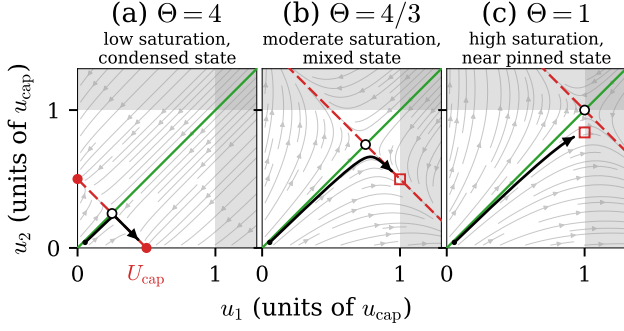


FIG. 3. **The pinning saddle across Θ .** Phase portraits of the two-schema strength dynamics Eq. (24) in (u_1, u_2) , units of u_{cap} , at three budgets $U_{\text{cap}} = Mu_{\text{cap}}/\Theta$ ($M = 2$ throughout): the pinned (saddle) point (black-edged circle), at $U_{\text{cap}}/2$, moves out along the stable manifold $u_1 = u_2$ (green) panel to panel as $U_{\text{cap}} \rightarrow Mu_{\text{cap}}$. The black curve is a trajectory from one seeded start (black dot, the same in all three panels), which grows along the stable manifold and then departs along the unstable one. (a) $\Theta = 4$: the budget alone halts growth well inside u_{cap} , so the unstable manifold (red, condensation) runs to a genuinely reached bare condensate (filled circles). (b) $\Theta = 4/3$: the ceiling is now reached first, arresting the same departure at u_{cap} (red square) before it reaches the excluded region $u_1 > u_{\text{cap}}$ or $u_2 > u_{\text{cap}}$ (shaded). (c) $\Theta = 1$: capacity exactly meets demand ($M_{\text{cap}} = M$), so the saddle sits at $u_1 = u_2 = u_{\text{cap}}$ and the trajectory is arrested (red square) just short of it. Panel sub-titles name the resulting state.

furcates as a single avalanche (class i). (b) The *spiked* state: one condensate holding the budget with a fringe pinned at the invasion threshold, reached when the budget cannot cap even one schema (class v). Θ exchanges them—the degenerate cluster shrinks from M schemas ($\Theta \leq 1$) to M_{cap} ($1 < \Theta \leq M$) to one ($\Theta > M$, the spike).

Leaving the saddle. The dynamics leave the saddle one axis at a time, along the engine chain *schema* $\rightarrow$ *differentiation* $\rightarrow$ *signal* $\rightarrow$ *coordination* $\rightarrow$ *resources*. An inactive axis sits with its covariance eigenvalue at the noise floor $\sigma_\omega^2/2\gamma$. When an axis activates, the resources it generates reweight $\mathbf{C}_r$ toward the agents who differentiated along it. Because their identities also spread in other directions, this inflates the residual variance off the active axis, and the next schema grows until its eigenvalue clears the floor and ignites. Successive axes leave the floor in turn, and this sequence is the endogenous cascade.

B. The regime map and its exact boundaries

Sequential sort and the two emergent rates. The candidate schemas do not cross threshold simultaneously: near threshold the λ_m of the growing cohort are nearly equal, so the common growth rate converts the seeded spread in

u_m into a staggered sequence of threshold crossings. This transient, which winnows and orders the seeded pool into a strength hierarchy, is the *sort*. Two rates characterize it. The *compounding rate* is read from Eq. (24) for an active, pre-saturation schema (where the channel is slack and $\varphi \approx 0$),

$$r_{\text{comp}} \equiv 2(\alpha\lambda_\star - \kappa), \quad (25)$$

$\lambda_\star$ the common resource-weighted variance of the growing cohort. r_{comp} is set by the gain-spectrum gap at the growing axis. The *activation rate* r_{act} is the rate at which successive survivors cross threshold. The decay and the budget pressure are common to every schema, so they cancel from the log-differences, $d\ln(u_m/u_n)/dt = 2\alpha(\lambda_m - \lambda_n)$. The common part translates the spectrum rigidly in $\ln \mu$ and moves every threshold together. Only the differences in λ_m reshape it, so the log-spacings hold while the growing cohort's λ_m stay near-equal. This is the quasistatic spectrum of the classes of Sec. III assume, and λ_m increasing with u_m eventually breaks it. Candidates seeded as in Sec. I compound at the common rate r_{comp} and so reach a fixed threshold at times spread by $\Delta \ln u/r_{\text{comp}}$, giving

$$r_{\text{act}} \propto r_{\text{comp}}/\sigma_u, \quad (26)$$

The readout $r \equiv r_{\text{act}}/r_{\text{comp}} \propto 1/\sigma_u$ is therefore fixed by the dispersion of the candidate repertoire and is not set independently. Both r_{act} and r_{comp} are emergent. σ_u is a property of the repertoire, which for a deployed system is operator-held (Appendix C).

The geometric and Zipf classes are morphologies of the sequential sort toward those endpoints, not fixed points. A monitor watches roles form, so these morphologies are the observable object. Slow age-ordered relaxation is geometric (class ii, $k_0 = r_{\text{act}}/r_{\text{comp}}$), a share-proportional sort under a budget at capacity is Yule–Simon and hence Zipf (class iii, $a \sim r_{\text{comp}}/r_{\text{act}}$), and an unorganized exogenous repertoire is Marchenko–Pastur (class iv). The replicator thus selects the cascade in two steps: Θ fixes the endpoint, and $r = r_{\text{act}}/r_{\text{comp}}$ fixes the transient morphology and its lifetime. No spectrum is imposed.

The cascade morphology depends on ~ 5 dimensionless groups, but only two separate regimes: the schemas per capacity slot $\Theta = M/M_{\text{cap}}$ and the activations per compounding time $r = r_{\text{act}}/r_{\text{comp}}$. The capacity M_{cap} is distinct from the realized survivor count K , since survivors need not be saturated. The two also have different ranges: M_{cap} exceeds M in the slack regime ($\Theta < 1$) and drops below 1 in the winner-take-all regime ($\Theta > M$), whereas $K \leq d$.

No intrinsic ceiling. Neither ceiling is intrinsic: the bare reinforcement supplies no turnover of its own. Along a single active axis the thick-phase strength is $\lambda(u) = \Lambda u - 1 + \sigma_\omega^2/4\gamma(\Lambda u - 1)$ (condensate amplitude plus fluctuation correction), which grows without turning over: $d^2\lambda/du^2 = \Lambda^2\sigma_\omega^2/2\gamma(\Lambda u - 1)^3 > 0$ throughout the supercritical range, so no *intrinsic* strength ex-

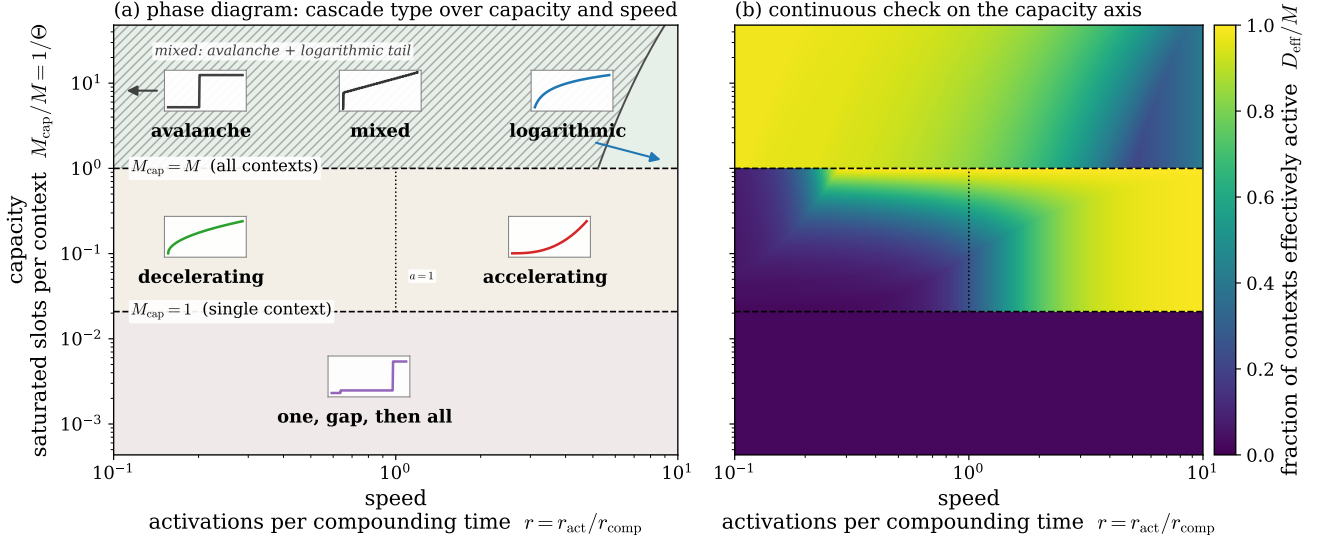


FIG. 4. **Cascade types and repertoire size over replicator dynamics.** Endogenous cascade regimes over the two coordinates that separate them. The horizontal axis (*speed*) is $r = r_{\text{act}}/r_{\text{comp}}$, the number of axes that activate per compounding time $1/r_{\text{comp}}$, fixed by the dispersion of the seeded repertoire through Eq. (26). The vertical axis (*capacity*) is M_{cap}/M , the saturated slots per context, that is the fraction of the M candidate schemas the finite social capacity can sustain as live contexts ($M_{\text{cap}} = U_{\text{cap}}/u_{\text{cap}}$). **(a) Phase diagram:** where each cascade type sits in the plane. The capacity coordinate alone, $\Theta = M/M_{\text{cap}}$, splits it into three bands with sharp counting-threshold boundaries, the contexts the society *can* sustain being all M , a graded cluster of M_{cap} , or one. Inset sketches show the sequence of activations the transient sort passes through, set by r : avalanche to logarithmic in the thick band ($\Theta \leq 1$), decelerating ($a > 1$) to accelerating ($a < 1$) across $a = 1$ in the middle band, and one onset, a gap, then all the rest in the thin band. The left-hand arrow marks $r \rightarrow 0$, where the cascade is a pure avalanche. The random Marčenko–Pastur class (iv of Fig. 2) is absent because the loop selects endpoints, and an unstructured random spectrum is an initial condition rather than an endpoint (Sec. IV B). Within the thick band a saturated cluster coexists with a geometric tail (hatched) below the exact boundary $r = M/L(\Theta)$. The right-hand arrow runs from the logarithmic sketch past that boundary to a purely geometric spectrum, and the inset shows $N(\beta)$ at a coexistence point—an early jump as the cluster ignites at once, then the tail’s gradual rise. **(b) Continuous check on the capacity axis:** the contexts *effectively active*, the participation ratio $D_{\text{eff}} = (\sum_k \mu_k)^2 / \sum_k \mu_k^2$ of the spectrum $\{\mu_k = u_k\}$, normalized by M , a continuous counterpart to the count M_{cap}/M on the vertical axis. It depends on how far the transient sort has progressed, hence on r , and falls below capacity wherever the sort is incomplete—bright on the accelerating side, darker on the decelerating side. The difference between the panels measures distance from the stationary endpoint. Exact boundaries, their mean-field status, the ceiling u_{cap} , and the held-fixed groups ($\alpha\langle\lambda\rangle/\kappa$, $\beta\mu_1/\gamma$, M/d) in Sec. IV B.

ists at which reinforcement saturates. The quartic confinement of Eq. (2) caps the differentiation *amplitude* ($m^2 = \beta\mu/\gamma - 1$) but not the schema *strength*, which keeps rising as the two role modes separate.

Two imposed ceilings. Two imposed ceilings arrest the growth of u , and they are distinct: a per-schema cap u_{cap} , and the total budget U_{cap} enforced by the soft pressure $\varphi(U)$ of Eq. (24). We take u_{cap} schema-independent. Both are properties of the legible-status channel of Eq. (4): its finite power limits U , its finite range caps any single u_m . That independence leaves Θ free. A per-axis share of one budget, $u_{\text{cap}} = U_{\text{cap}}/M$, would fix $\Theta = 1$ and remove the coordinate. The channel’s third number is its noise scale σ_{read} (the measurement-quality limit of Ref. [17]), the finest status difference it resolves. $M_{\text{cap}} = U_{\text{cap}}/u_{\text{cap}}$ counts the axes that can be driven to full range at once, and Θ is demand measured against M_{cap} . The channel-to-strength mapping rests on a convention, so this fixes u_{cap} up to proportionality, which is

all Θ requires.

Reading the map. Figure 4 is the principal (Θ, r) slice, a phase diagram locating the cascade types (Fig. 4a) with a continuous check on its count-based capacity axis (Fig. 4b). The five spectra of Fig. 2, all at $R = 50$, already span D_{eff}/M from 0.05 (Zipf $a = 2$) to 1 (degenerate), so the realization panel reads out cascade class and not capacity alone. The two coordinates act on different objects. Θ selects the stationary endpoint: the saturation ceiling caps the condensation instability, in which all strength collects on one schema, when capacity exceeds demand ($\Theta < 1$, giving a degenerate spectral cluster), and leaves that instability unchecked when the budget cannot sustain even one schema ($\Theta > M$, giving a spiked spectrum). Its boundaries are therefore sharp phase lines. r measures how far apart in time successive candidates cross threshold. When the seeded repertoire is widely dispersed, the earliest contexts compound longest and incumbents entrench; when it is nar-

row, candidates ignite together and churn instead. The two limits are a smooth crossover between the geometric and Zipf classes. Here $r = r_{\text{act}}/r_{\text{comp}}$ is the generative shape parameter fixed by the seed dispersion through Eq. (26). Four endogenous regimes arise. An early ceiling re-equalizes the winners into a *saturated cluster* ($\Theta \leq 1$), near-degenerate, class (i). A standing cohort encodes age as strength, $u_m \propto e^{r_{\text{comp}}(t-t_m)}$, giving the *geometric* regime ($r \ll 1$), class (ii). A budget at capacity grows shares in proportion, a Yule–Simon process [35] giving *Zipf* ($1 < \Theta \leq M$), class (iii). Unregulated condensation leaves one winner over a fringe pinned at the invasion threshold, the *spiked* regime ($\Theta > M$), class (v). Figure 4a places them in the plane.

Scope. The map takes the repertoire near-orthogonal, parametrizes capacity by Θ alone, and describes a transient rather than a stationary hierarchy; [26, Sec. S7] sets out what each of these assumes and where it fails.

Boundaries. Three of the four boundaries are counting thresholds in Θ and one is a threshold in r , and all four are exact analytic curves on the principal slice (Appendix E).

V. MONITORING: READING THE CASCADE BEFORE IT STARTS

Consider a monitor who observes what the agents do and nothing else. As the loop gain rises, it sees role axes activate one at a time, and each activation appears as a split in the population’s behavior along one such axis. The monitor’s task is to forecast: it needs a quantity, measurable before the next activation, that predicts the rest of the sequence, ideally with enough precision to determine the cascade class. The eigenvalues of the behavioral covariance $\mathbf{C}$ (Eq. (13)) are such a quantity. A class is a shape of the gain spectrum ρ , and Eq. (20) makes N depend on the products $\Lambda\mu_k$ alone, so a mode’s onset sits at $\ln \Lambda = -\ln \mu_k$. An error $\delta \ln \mu$ common to the spectrum therefore displaces every predicted onset by $-\delta \ln \mu$ in $\ln \Lambda$. Over a lead time Δt the forecast is in error by

$$\delta N \simeq \underbrace{\frac{dN}{d \ln \beta}}_{\text{bifurcation density}} \left(\underbrace{\delta \ln \mu}_{\text{spectrum as measured}} + \underbrace{r_{\text{comp}} \Delta t}_{\text{drift over lead time}} \right).$$

This bifurcation density, the number of onsets per unit $\ln \beta$, is $dN/d \ln \Lambda = R\mu\rho(\mu)$ (Eq. (22)). It converts a spectral error into a class error: where onsets are dense, a given $\delta \ln \mu$ blurs more of them together. The two error terms are properties of the monitor’s situation, and the density is a property of the repertoire.

The measurement term depends on how $\{\Lambda_k\}$ is estimated, and there are two routes. One is the relaxation time of the k -th mode of $\mathbf{C}$, $\tau_k = 1/\gamma|1 - \Lambda_k|$, which requires observing for $O(\tau_k)$, and τ_k diverges at threshold. This is the critical slowing down whose rise is a

generic early-warning signal [36]. If the engine parameter is swept at rate $\dot{\Lambda}$, the dimensionless ratio $\mathcal{R} = \tau_k \dot{\Lambda}$ controls feasibility. Only the local rate enters, so a sweep of any smooth shape is covered by its slope at the crossing, and an adoption curve is worst for a monitor where it is steepest. When the forcing is fast relative to the system’s internal timescale, the sweep crosses the transition before the slowing can be measured. This is rate-induced tipping, where slowing-down indicators generically fail [37, 38]. Slow deployment ($\mathcal{R} \ll 1$) preserves genuine early warning, and fast deployment forecloses it.

The other route reads the same quantity from the covariance. Subcritically $\mathbf{C} = \frac{\sigma_\omega^2}{2\gamma}(\mathbf{I}_{d \times d} - \mathbf{\Lambda})^{-1}$, so $v_k = \frac{\sigma_\omega^2}{2} \tau_k$: the k -th covariance eigenvalue is the k -th relaxation time. That identification uses the flat noise floor: at finite encounter rate the shot noise of Appendix A adds a term proportional to μ_k to the numerator, so v_k overstates τ_k on the strongest modes and biases the recovered Λ_k upward there. A sample of agents at one instant estimates it with no time record, and the modes still to activate sit far below threshold, where $\tau_k \approx 1/\gamma$ and the population has long since equilibrated. This route is limited by the number of agents rather than by the observation time. Sampling fluctuations give each measured eigenvalue a finite width, and eigenvalues closer together than that width cannot be separated. Inverting the empirical eigenvalue distribution of behavioral embeddings therefore recovers $\rho(\mu)$, and with it the full $N(\beta)$ (Fig. 2a), its bifurcation rate (Fig. 2b), and any avalanche masked by an early onset (Fig. 5b). Rate extrapolation recovers none of this, since it samples ρ only at $\mu \geq \gamma/\beta$ and must assume ζ below (Eq. (23)). Where several onsets have already been observed, their spacings identify the class (Sec. V A). Before the first onset, and inside a quiescent interval, no spacings exist to fit, so only the inversion applies.

The drift term grows with the lead time. The eigenvectors of $\mathbf{\Lambda}$ are the context directions, which the loop holds fixed, and its eigenvalues follow the schema strengths, which compound at r_{comp} . A drift common to every mode shifts $\ln \mu$ uniformly, which displaces the cascade in β and leaves its class unchanged. Only the spread of r_{comp} across surviving modes changes the class, so the class forecast outlives the timing forecast. Both hold over $\Delta t \lesssim 1/r_{\text{comp}}$, long whenever the schemas drift slowly compared with the sweep, since r_{comp} is of order κ . Where the repertoire is not near-orthogonal the eigenvectors move with the strengths as well, and that limits the forecast instead (Sec. IV B).

A. Consequences for monitoring and design

The bifurcation density decides what either error costs, and unlike them it is a property of the repertoire rather than of the monitor. It is largest where the spectrum is dense, so clustered onsets blur together under a spec-

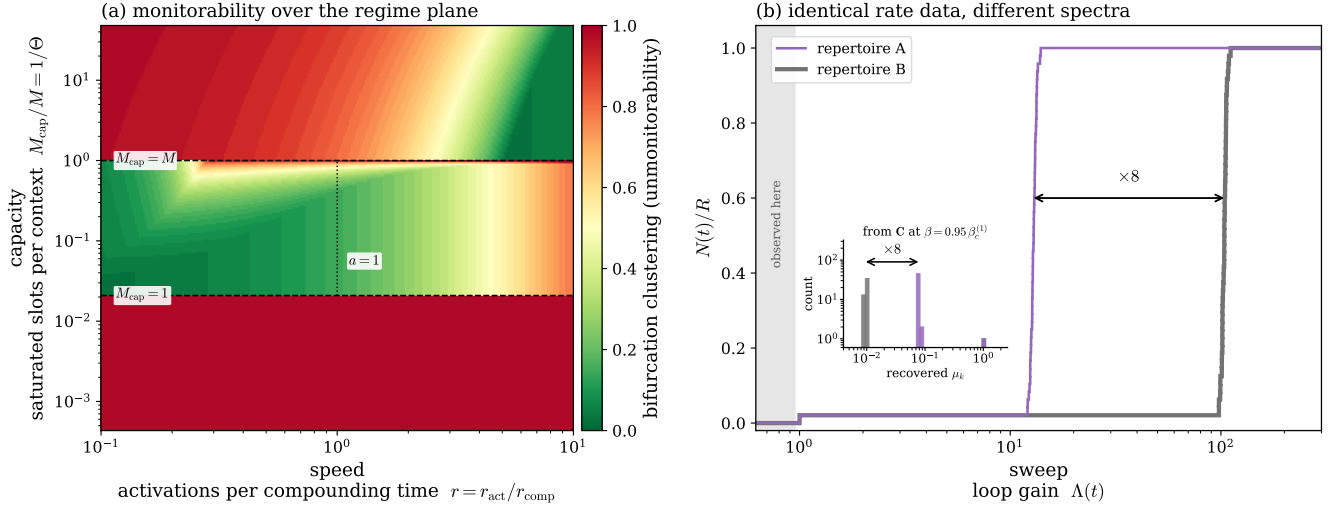


FIG. 5. **Monitoring the cascade.** (a) Monitorability over the (Θ, r) plane: the clustering of bifurcation onsets (colour; the maximal share arriving within $\Delta \ln \beta = 0.15$, hence unmonitorability), red where onsets cluster into a single burst (avalanche, spiked) and green where they resolve into a sequence (logarithmic, decelerating). The Θ boundaries are sharp, the r dependence a smooth crossover. Regime layout as in Fig. 4. (b) The measurement term, under a linear sweep of the loop gain $\Lambda(t)$. Two repertoires whose counting functions coincide over the whole plotted range: each activates one axis at $\Lambda = 1$ and then stays quiescent. In repertoire A the remaining $R - 1$ modes sit just below threshold and avalanche at $\Lambda = 13$; in repertoire B the same modes sit far below and do not activate until $\Lambda = 104$. The two spectra differ only below $1/\Lambda$, which the bifurcation rate never samples, so rate extrapolation cannot separate them. Inset: the gain spectrum recovered from the subcritical covariance $\mathbf{C} = \frac{3}{2\gamma} (\mathbf{I}_{d \times d} - \Lambda)^{-1}$ at $\Lambda = 0.95$, before either has bifurcated. The two recovered bulks are separated by the same factor of 8 that separates the two rises, since an onset sits at $1/\mu$. The paired arrows mark it. The inversion amplifies a 7% difference in covariance eigenvalues into that factor, so the number of agents sampled sets the practical limit (Sec. V).

tral error and the classes stop being separable. Figure 5a maps it over the (Θ, r) plane as the largest share of onsets arriving within a fixed window in $\ln \beta$. An operator who reshapes the spectrum therefore changes how monitorable the system is, whatever the monitor does.

The classification converts a low observed rate into a decision procedure. The first few inter-bifurcation spacings identify the class: constant ratios β_{k+1}/β_k indicate geometric (estimate k_0 , predict the rest); ratios growing as $((k+1)/k)^a$ indicate Zipf (the exponent's position relative to $a = 1$ decides decay versus explosion); a rate rising as $(\Lambda - 1)^{1/2}$ indicates an imminent random bulk. A long quiescent interval is not evidence that the cascade has ended, since it is also the spiked signature. Rate data alone cannot distinguish the two, and the subcritical spectral inversion resolves the ambiguity. The complementary design levers on the cascade class are collected in Appendix C.

The class structure also makes the cascade designable, but the prescription is not monotone in $r = r_{\text{act}}/r_{\text{comp}}$: it reverses between capacity bands. Where the budget caps growth ($1 < \Theta \leq M$) the sort is Yule-Simon with exponent $a = 1/r$, so a widely dispersed repertoire (lower r) steepens the schema-strength hierarchy, each activated schema compounding longer before the next crosses threshold. That spreads the bifurcation thresholds into the extrapolable regime. A narrow reper-

toire (higher r) flattens spectra and moves the system toward the avalanche-prone classes. In the thick band ($\Theta \leq 1$) the operative boundary is instead $r = M/L(\Theta)$ (Sec. IV B): below it a saturated cluster ignites at once, so there the preference reverses and it is larger r that resolves the cascade into a sequence. Independently of r , per-context pairing efficiencies $\phi_m = 1 - \rho_m$ reweight the effective spectrum $\mathbf{H}^\top \text{diag}(\phi) \mathbf{H}$, letting an operator stretch clustered eigenvalues into a resolvable sequence at some cost in coordination efficiency. These levers also frame a conjecture about deployment, in which parameters Θ and r (fixed here) might evolve systematically, moving the system toward the avalanche-prone corner of Fig. 5: more mature systems sitting at higher Θ (more coordination competing for finite capacity) and, as a platform's task taxonomy fills in around a settled core, at higher r (a narrower spread of context weights). Engineering that improves coordination would then erode its own monitorability. Engineering practice outside the present theory sets the drift of these parameters across a system's lifetime. Making that drift an endogenous trajectory would require promoting Θ and r to slow dynamical variables, a two-timescale extension we leave open.

VI. DISCUSSION

Distributional AGI: a control surface. Recent safety work [25] hypothesizes that general-level AI capability may emerge not in one model but from the coordination of many sub-general agents. Our framework makes this quantitative: the thick phase is the coordinated collective. Shared substrates—task queues, codebases, finite compute—make same-action encounters interfere, which gives the payoffs an anti-coordination structure. Complementary roles (planner/executor, coder/reviewer) then extract gains no member could achieve alone, sustained by self-generated signals rather than assigned by an orchestrator. There is growing evidence of the benefits of coordination for tasks assigned to groups of agents [10].

Two consequences follow for the distributional AGI hypothesis. First, an operator can cross the transition by improving the coordination layer alone—routing, protocols, channel noise ($\rho_{\text{pair}}, \sigma_{\text{read}}$)—with no change in any agent, so per-agent capability evaluations will not detect it. At scale the crossover sharpens toward a discontinuity. Second, orchestration cannot compensate below the cognitive floor. Agent populations with $\eta_{\text{agent}} < \sigma_{\text{read}}\gamma/4\beta$ cannot ignite the information engine that runs the role-based phase, and inference quality caps the number of sustainable roles through the dense-associative-memory bound $M_{\text{eff}} \leq d^{n-1}$ [29], with n (policy sharpness) rising with capability [39]. Crude agents sustain a coder and a reviewer, not an organization, but the ceiling d^{n-1} is exponential in n : one more unit of policy sharpness multiplies it by d ([26, Eq. (S12)]).

Steering acts on the same object from outside. Character traits in a language model can be monitored and controlled along directions in its activation space [40]. The loop reaches those directions from within. An agent’s identity ω_i accumulates its own past role-taking and re-enters its context through memory, so the population’s coordination history displaces each agent much as an imposed steering vector would. The theory adds a stability criterion: below ignition an imposed separation relaxes back to the noise floor, and above it the population holds the separation with no steering at all, because the separation induces roles that pay. The monitor of Sec. V is the population-level counterpart of the same measurement, projecting behavior onto role axes where persona work projects activations onto a trait direction.

The same factors that set ignition and shape the cascade are *platform-owned* apart from the agent capacity η_{agent} , so an operator who fixes the environment before ignition holds a control surface over which takeoff scenario is reachable. It is a design choice among settings, not control-theoretic steering during takeoff, which the rate-induced-tipping limit forbids (Fig. 5). We set out the available levers, their costs, and the two tensions that limit how far they can be pushed in Appendix C.

Outlook. The model realizes Sewell’s duality as a fixed-point equation—schema $\leftrightarrow$ context direction, resource $\leftrightarrow$ accumulated coordination payoff, structure $\leftrightarrow$ resource-

weighted covariance, agency $\leftrightarrow$ deviation from prescribed role, transposability $\leftrightarrow$ context overlap $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n$. One object, the spectrum of the social gain matrix, then organizes onset, cascade shape, monitorability, and design. The main scope condition concerns transformation: the strengths u_m are endogenous, the axes $\hat{\mathbf{g}}_m$ are not. The loop therefore goes beyond reproduction, since which schemas govern is itself an outcome. It stops short of transformation in Sewell’s sense, which requires a schema to change what it is about. Sewell attributes social novelty to that transformation. Part of that mechanism is already present as the overlap $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n$, through which differentiation along one axis raises the resource-weighted variance along another. Between orthogonal contexts the coupling is competitive only, through the shared budget. Moving a schema onto a new axis requires the angular sector the closed loop does not evolve, and Appendix D sets out what an extension of it must decide. Nor does the model capture Sewell’s claim that transformation is perpetual: our dynamics sort into roles and settle to a pinned or condensed fixed point, whereas a Sewellian society never settles. Sustained schema multiplicity lies outside the closed loop. We leave it open.

Two extensions remain within the emergence regime. Retaining the role-sign dependence of coordination income, which the mean field averages away, lets the game’s payoff asymmetry (6 versus 2) concentrate resources in dominant roles. That adds endogenous stratification to differentiation. A mixed repertoire of anti-coordination and conformity games couples the differentiation sector treated here to a consensus sector, with gain eigenvalues of both signs. Role axes inflate the covariance, consensus axes shift the mean, and the covariance-spectrum monitor sees only the former. The most concrete near-term task is to test the mean-field predictions in finite-population simulation, where fluctuations select which schema condenses. Two smaller questions follow: whether resource weighting generically widens spectral gaps, which would enlarge the monitoring window, and whether truthful status presentation can be made an equilibrium of a strategic-display extension [41].

For informational active matter the engine here supplies a second kind of measurement-induced order. Existing realizations reach consensus, whose order parameter is a mean [22, 23]. Differentiation orders the covariance instead, so the transition is a cascade indexed by the eigenvalues of the social gain matrix rather than a single bifurcation. Two features carry beyond this model. The gain matrix is itself dynamical, so the spectrum that sets the transitions is an output rather than an input. And the subcritical covariance is the static response to that gain, so inverting it recovers the spectrum before any transition occurs, a readout available wherever the interaction is a matrix rather than a scalar. The bound of Sec. I is stated in information units, as is the consensus bound it parallels [22]. For the engine here the reason is that the mean-field reduction averages the action over the observation noise, and the resulting gra-

dient dynamics produces no entropy. Carrying the accounting at the level of the individual encounter, where the measurement record still exists, would put the pay-off and the cost of that record in the same units, and is the step measurement-induced order needs to become thermodynamic. The loop here is autonomous rather than cyclic, so that accounting runs through the information flow between coupled subsystems [42] rather than through a measurement-and-reset cycle. Refs. [43, 44] carry the accounting out for a single agent operating a cyclic engine, subtracting the cost of the memory from the work its inference yields. The strategy that maximizes gain alone is not the one that maximizes that net output. Recent achievements in information engines have come chiefly from experimental realizations [20, 45], in pursuit of grounding them further in stochastic thermodynamics [16]. Populations of AI agents supply an analogous experimental system for the collective case treated here. The platform sets their parameters rather than a physical substrate, so an experiment can be built to the theory. Extending that accounting from the single agent to the population, and measuring it on such a platform, is what an information thermodynamics of agent populations would consist of [15].

Populations of AI agents are the setting in which the loop is testable. The parameters a human society only permits us to infer are set there by the platform. Agent identity ω_i is an embedding of the persistent memory the system keeps for each agent, and a context direction $\hat{\mathbf{g}}_m$ is a task type in which two agents take complementary roles. The write rate β and the decay γ are the weight with which encounter outcomes enter that memory and its retention half-life, and the channel factors σ_{read} and ρ_{pair} are the visibility of a partner’s standing and the router’s matching policy (Appendix C). A sweep of Λ through any of them therefore tests the ignition condition directly. An operator can also locate a deployment relative to threshold from a single-agent evaluation of η_{agent} and a measurement of the channel, without running it supercritical ([26, Sec. S2 C]). The cascade itself is read by embedding behavior and counting the eigenvalues of the population covariance $\mathbf{C}$ above the noise floor, and the first few spacings classify it (Sec. V). Role differentiation that appears without the subcritical covariance signature would indicate a mechanism other than the one proposed here.

The same setting can settle modeling choices the theory now fixes by assumption. The identity noise is the clearest case. We take it isotropic and independent of the drive, which gives the cascade a common floor to be read against. The encounter shot noise instead aligns with the context directions and grows as β^2/ν (Appendix A). The ignition thresholds $\Lambda_k = 1$ follow from the drift and hold under either choice. The flat floor, the subcritical inversion of Sec. V, and the separation between invasion and ignition depend on it. A platform logs the encounter statistics and the identity dispersion together, so which noise dominates is open to measurement. The repertoire

geometry and the closed pool admit the same treatment. The theory supplies the consequences that follow once such assumptions are fixed, and fixing them empirically is where a science of agent populations would begin.

The formulation is deliberately minimal: one anti-coordination game, a Gaussian mean field, a loop that settles rather than perpetually transforms. It describes no real society, but serves as an idealization in the spirit of the harmonic oscillator—solvable, built around a single mechanism, and extended by perturbation. Here that mechanism is information-fuelled role differentiation and the cascade that follows. Whether the spectrum of a social gain matrix proves as useful across the social sciences as the oscillator’s spectrum is across physics is the question we mean to raise.

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DATA AVAILABILITY

The simulation and figure-generation code that supports the findings of this study is openly available at https://github.com/mptouzel/role_diff. No experimental datasets were generated or analyzed.

Appendix A: Microfoundations of the mean-field dynamics

The main-text role dynamics is the mean-field drift of an underlying encounter process. We record the reductions that produce it, since each is a modeling assumption with its own scope.

Encounter clock and the write rate. Agent i plays at Poisson times with rate ν . At an encounter in context m against partner j it commits a signed decision $a \in \{-1, +1\}$ and inscribes a fixed increment $b_0 a \mathbf{g}_m$ into ω_i , which otherwise relaxes:

$$d\omega_i = -\gamma \omega_i (1 + |\omega_i|^2) dt + b_0 \sum_m \mathbf{g}_m dN_{im}(t) + \sigma_\omega d\mathbf{B}_i, \quad (\text{A1})$$

with N_{im} a counting process of rate νf_m with signed jumps a . Taking the compensator, $\mathbb{E}[dN_{im}] = \nu f_m \langle a \rangle_m(\omega_i) dt$, so the drift depends on b_0 and ν only through the product $\beta \equiv \nu b_0$: encounter frequency and per-encounter inscription are not separately identifiable at mean field. This replacement is the window average of Sec. I written as a differential. It discards a martingale with increments of order $\sqrt{\nu f_m dt}$, the shot noise quantified below. The same ν multiplies the coordination income (Sec. II B), so setting $\nu = 1$ (time measured in average inter-encounter intervals) is a global units choice,

not a per-equation one. $\beta/\gamma = \nu b_0/\gamma$ is then the number of times an identity is rewritten within its own decay horizon.

Order of the two limits. The window average of Sec. I carries out the sum over partners on its own. At fixed N_a the drift is $\beta \sum_m f_m \mathbf{g}_m \sum_j p_{ij}^{(m)} \bar{a}((\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m)$ with $\bar{a}$ the noise-averaged action of Eq. (A2), so the pairing probabilities already enter as weights. That equation is not closed, since its right-hand side carries the other $N_a - 1$ identities. The limit $N_a \rightarrow \infty$ closes it. It removes the imprint of the focal agent on its own partners, which is $O(1/N_a)$ because ν is a fixed per-agent budget, and the run-to-run spread of the sum over the realized identities. The pairing average therefore follows from dense encounters alone ([26, Sec. S8]).

Signed decision and its noise average. At the decision level the action is a hard sign of the integrated evidence, $a = \text{sign}(\frac{1}{T} \sum_t s_t)$ on samples $s_t = \Delta + \sigma_{\text{read}} \xi_t$ of the true status difference $\Delta = (\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m$. What enters the drift is its expectation over the observation noise,

$$\bar{a}(\Delta) = \mathbb{E}_\xi [\text{sign}(\bar{s}_T)] = \text{erf}\left(\frac{\sqrt{T} \Delta}{\sqrt{2} \sigma_{\text{read}}}\right), \quad (\text{A2})$$

$$\bar{a}'(0) = g, \quad \bar{\pi}'(0) = \frac{\sqrt{T/2\pi}}{\sigma_{\text{read}}}. \quad (\text{A3})$$

The decision rule is a function of the observation s . The mean-field response is a function of the true status difference Δ , the noise having been integrated out. Averaging $\bar{a}$ over the partner in turn gives $\langle a \rangle_m(\boldsymbol{\omega}_i)$, the response that enters Eq. (2). The two symbols differ by that second average. Both are instances of one operation: smoothing an erf by a Gaussian returns an erf, with the two variances added in quadrature inside its argument ([26, Eq. (S25)]). Taken over the observation noise it produces Eq. (A2), and taken over the partner it softens the slope by the factor of [26, Sec. S6]. It is this slope $\bar{\pi}'(0)$, not the hard sign, that sets the gain: an un-averaged sign would give a singular (delta-function) susceptibility and an ill-posed threshold. Integrating T samples sharpens the response as $\sqrt{T}$ (the read memory), whereas the identity retention $1/\gamma$ is the write memory. The two are distinct stages of the cycle.

Context normalization. One context is drawn per encounter, with frequencies normalized so that $\sum_m f_m = 1$. The drift $\beta \sum_m f_m \langle a \rangle_m(\boldsymbol{\omega}_i) \mathbf{g}_m$ is then an expectation over which context is played rather than a sum that grows with M , and the gain operator $A = \sum_m f_m \mathbf{g}_m \mathbf{g}_m^\top$ has $\text{tr } A = \sum_m f_m u_m$, which stays $O(1)$ as contexts are added provided the capacity grows with the repertoire ($U_{\text{cap}} \propto M$, i.e. fixed Θ of Sec. IV B) rather than being held fixed: adding contexts at fixed budget divides the same write rate more finely rather than driving identity harder. This read budget $\sum_m f_m = 1$ is the counterpart on the read side of the capacity budget $\sum_m u_m \leq U_{\text{cap}}$ on the write side (Sec. IV A). The dependence on how many contexts probe a d -dimensional space enters through the aspect ratio M/d (Sec. III), not through an overall scale.

Pairing kernel and the well-mixed reduction. Partners are drawn with probabilities $p_{ij}^{(m)}$ ($\sum_j p_{ij}^{(m)} = 1$), collected in the pairing kernel $\mathbf{P}^{(m)}$. Linearizing the partner-averaged response near the symmetric state,

$$\sum_j p_{ij}^{(m)} \bar{a}((\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m) \approx g \mathbf{g}_m^\top ((\mathbf{I}_{N_a \times N_a} - \mathbf{P}^{(m)}) \boldsymbol{\Omega})_i, \quad (\text{A4})$$

with $\boldsymbol{\Omega}$ the N_a identities stacked, so pairing enters the gain through the graph Laplacian $\mathbf{L}^{(m)} = \mathbf{I}_{N_a \times N_a} - \mathbf{P}^{(m)}$ of the interaction network, and ignition is governed by the spectrum of $\sum_m f_m \mathbf{g}_m \mathbf{g}_m^\top \otimes \mathbf{L}^{(m)}$ —a coupling of cultural space to social space. For *well-mixed* pairing $p_{ij} = 1/N_a$, $\mathbf{P} = \frac{1}{N_a} \mathbf{1}\mathbf{1}^\top$ projects onto the population mean, $\mathbf{L}$ acts as the identity off that mean, and the partner-average reduces to coupling to $\langle \boldsymbol{\omega} \rangle$ (a Curie–Weiss field). This recovers $\boldsymbol{\Lambda} = \boldsymbol{\Lambda} \mathbf{H}^\top \mathbf{H}$ and a scalar pairing factor: uniform p_{ij} gives $\rho_{\text{pair}} = 0$ and factor 1, while an annealed matcher that conditions on status gives the general $(1 - \rho_{\text{pair}})$ of [26, Sec. S2 B] (the uniform diagonal reweighting of [26, Sec. S2]). The results of this paper are the annealed case. Structured p_{ij} —homophily, communities, degree heterogeneity—makes $\mathbf{L}^{(m)}$ nontrivial. It shifts thresholds and lets network structure, not only $\|\mathbf{g}_m\|$, select which role axis ignites first. Because homophily makes p_{ij} depend on $\{\boldsymbol{\omega}_i\}$, the interaction network then co-evolves with the culture. What we bracket is that quenched structure, not status-conditioned matching: an annealed matcher leaves no network to accumulate, so it decouples network from culture and closes the loop analytically.

Shot noise. At finite ν the compensated jump stream contributes shot noise of intensity $\nu b_0^2 \sum_m f_m \mathbf{g}_m \mathbf{g}_m^\top = \beta b_0 \mathbf{H}^\top \text{diag}(\mathbf{f}) \mathbf{H}$. This is proportional to $\mathbf{H}^\top \mathbf{H}$ rather than isotropic, so it is aligned with the context directions and largest along the active axes. It is not a scalar renormalization of σ_ω : retaining it replaces the flat floor $\sigma_\omega^2/2\gamma$ by a mode-dependent floor that grows with μ_k . The poles of the covariance are unchanged, because the shot-noise covariance shares the eigenbasis of $(\mathbf{I}_{d \times d} - \boldsymbol{\Lambda})^{-1}$ and reweights only its numerator. The thresholds $\Lambda_k = 1$ and the relaxation-time spectroscopy are therefore identical in the two cases, and only the pre-onset amplitudes differ. The intensity is $O(\beta b_0) = \beta^2/\nu$, so it grows with the drive rather than holding still as β sweeps, and the two noise sources are not interchangeable: σ_ω supplies a fixed isotropic reference and this one neither. It vanishes in the dense-encounter limit ($\nu \rightarrow \infty$, $b_0 \rightarrow 0$ at fixed β), which leaves the isotropic $\sigma_\omega^2 \mathbf{I}_{d \times d}$ that gives the flat noise floor used in Sec. III. The same granularity carries the model’s irreversibility: partners observe each other independently, so a miscoordinated encounter kicks both identities the same way, which no pairwise potential generates, and P_e therefore sets the coordination payoff and the entropy production together (Sec. VI). The floor is flat to the extent that $\beta b_0 = \beta^2/\nu \ll \sigma_\omega^2$, that is where a single encounter moves an identity by much less than σ_ω^2/β . We take this limit throughout.

Why the identity noise is kept. Dropping σ_ω and letting the shot noise above be the only stochasticity would be more economical, since that noise is the measurement error the Szilard bound already concerns. Three properties decide against it here. The noise along an axis is then proportional to μ_k , so it vanishes with the strength, $\lambda_m \rightarrow 0$ as $u_m \rightarrow 0$, and no schema can grow before role structure exists (Sec. IV A). The floor becomes mode-dependent, so the cascade loses the common reference its thresholds are read against. The floor also grows as β^2/ν , rising during the sweep a monitor watches. All three raise λ_m preferentially on the strongest axes, so retaining them would steepen the strength hierarchy and move the system toward the avalanche-prone classes. The isotropic choice is therefore the conservative one for Sec. V.

Status microfoundation. The signal $s_{im} = (\omega_i - \omega_j) \cdot \mathbf{g}_m + \sigma_{\text{read}}\xi$ is agnostic between a noisy read of a public status difference and exact self-knowledge minus a noisy read of the partner’s self-stored status. The mean-field results are identical. We adopt the second reading—each agent stores its own status and presents a noisy version—so that no public registry is presupposed. This keeps the emergence claim non-circular (a public ledger is itself an institution, analyzed as operator technology in [26, Sec. S4]) and matches the feedback term βa_{im} , which inscribes the agent’s own act. Presentation is assumed truthful. The anti-coordination payoffs penalize inflated display (feigned high status recruits deference until it collides with genuine high status at $(-1, -1)$), so honesty is a plausible equilibrium of a strategic-display extension we do not develop here.

Appendix B: Simulation details

Analytics. The coordination payoff $W = w_0(1 - 2P_e)^2$ follows from enumerating role-assignment errors in the Go/Defer game. The loop gain $\mathbf{\Lambda} = \mathbf{\Lambda} \mathbf{H}^\top \mathbf{H}$ follows from mean-field linearization of Eq. (1). The self-consistent fixed point follows from a timescale-separated (identities $\rightarrow$ resources $\rightarrow$ schemas) analysis, in which identities equilibrate to the Boltzmann distribution of the gradient dynamics and resource-weighted covariances are evaluated by Wick’s theorem in the Gaussian regime. The endogenous spectra follow from the resulting strength replicator. Full derivations are given in Secs. II A–III and [26].

Simulation (Fig. 1). $N_a = 2500$ agents, $d = M = 3$, Euler–Maruyama integration of the dense-encounter limit of Eq. (1) (one random pairwise matching per step, all M contexts played per encounter at $f_m = 1$, single-sample policy), $\gamma = 1$, $\sigma_{\text{read}} = 0.6$, $\sigma_\omega = 0.25$. The schema repertoire is held fixed, with near-orthogonal context directions and a hierarchical profile of strengths ($|\mathbf{g}_m| = 1.55, 1.1, 0.72$), so the covariance eigenvalues $v_k(\beta)$ resolve the cascade cleanly at $\Lambda\mu_k = 1$ with $\mu_k = u_k = |\mathbf{g}_k|^2$, and $\rho_{\text{pair}} = 0$ (factor 1) for random pairing. The protocol is quasi-static: at each β on a grid

we equilibrate (1.2×10^3 burn-in steps, warm-started from the previous β) and average the order parameter over a further 1.5×10^3 steps, so panels b,c are functions of β rather than of time. The endogenous selection of the cascade ordering under dynamic schemas is treated in Sec. IV A. The full dynamic-schema code is available in the repository given under Data availability.

Appendix C: Platform control surface

The ignition and cascade factors are platform-owned apart from the agent capacity η_{agent} , so an operator holds a pre-ignition control surface over which takeoff scenario is reachable (Fig. 5). The parameters an operator sets are the write rate β and the retention $1/\gamma$ of the identity memory, the channel factors σ_{read} and ρ_{pair} —the visibility of a partner’s standing and the matching policy—the context play frequencies $\mathbf{f}$ and pairing efficiencies ϕ , and the repertoire itself: its directions $\{\mathbf{g}_m\}$ and their seed dispersion σ_u . Everything else in the loop is an outcome. Three levers act cleanly on it. *Ignition* is monotone in the channel factors ρ_{pair} and σ_{read} . Every agent must clear the cognitive floor, so raising channel noise or degrading matching shuts the engine down whatever the agents’ quality. This lever acts entirely outside the models. Disassembly does not: removing individual agents leaves the resource-weighted inertia in place and the order stands, much as it does when spins are removed from a ferromagnet. *Cascade class* is steerable directly. The effective gain reweights the context second-moment matrix by each context’s play frequency times its pairing efficiency, $\mathbf{H}^\top \text{diag}(\mathbf{f}) \text{diag}(\phi) \mathbf{H}$, so per-context rate limits (the frequencies f_m) and matching quality (the efficiencies ϕ_m) are interchangeable levers. Optimizing that schedule splits a spiked spectrum into a geometric, extrapolable sequence ([26, Sec. S2 D]). In a representative repertoire it more than halves the log-spacing variance a monitor must resolve ($1.35 \rightarrow 0.58$) and costs roughly 40% of the coordination efficiency. The same tradeoff recurs across the levers below: monitorability is obtained at the expense of coordination efficiency. *The sort* is set by the repertoire’s seeding rather than by the ledger’s memory: $r = r_{\text{act}}/r_{\text{comp}} \propto 1/\sigma_u$ (Eq. (26)), so tiering task types by launch traffic, rather than opening them at equal weight, lowers r . Where the budget caps growth ($1 < \Theta \leq M$) this shifts the cascade off the avalanche side toward the extrapolable classes. In the thick band ($\Theta \leq 1$) the preference reverses at $r = M/L(\Theta)$, and a narrower seeding resolves the cascade instead (Sec. IV B). The operator defines the task taxonomy, so σ_u is platform-owned in the same sense as $\mathbf{f}$ and ϕ and acts on the same object, the spectrum of the effective gain ([26, Sec. S4 A]). Two tensions run through these levers. Legibility trades against lock-in: the widely dispersed repertoire that yields the extrapolable classes is also the one in which the earliest and most strongly seeded contexts compound longest into entrenched incumbency, so the third lever improves legi-

bility and deepens lock-in at once. A benign setting must also be *held* rather than reached. The closed loop evolves toward condensation without intervention (Sec. IV A), so the system returns to the avalanche-prone classes once the intervention is withdrawn. Which levers an operator holds at all depends on where identity is kept. A platform-maintained status ledger gives the operator β , γ , and the matching correlation ρ_{pair} together. If identity memory migrates into the agents' own stores, those levers pass to the collective, and that migration is itself an early and observable warning that control over the knobs is being lost ([26, Sec. S4]). The dependence also runs the other way. A platform that measures $\mathbf{C}$ recovers the role axes as its eigenvectors and the schema strengths from the subcritical inversion (Sec. V). The drift in Eq. (1) is a sum over realized actions, so a platform that logs encounters can integrate that equation in parallel and serve s from its own estimate of ω_i , which tracks the identity up to the unobserved identity noise $\sigma_\omega \eta_i$. Equation (3) does not follow, since its driver is a payoff rather than an action, and Eq. (4) inherits that, because the strengths follow the resource-weighted covariance. The identity apparatus is therefore platform-ownable up to that noise, and only the reward path stays with the agents.

Appendix D: Schema directions and the vector form

Equation (4) evolves schema strengths on fixed axes. The general object is the vector reinforcement

$$\dot{\mathbf{g}}_m = -\kappa \mathbf{g}_m + \alpha \mathbf{C}_r \mathbf{g}_m, \quad (\text{D1})$$

which has the same two ingredients—covariance-driven and resource-weighted—but also lets a schema change which axis it prescribes. Writing $\mathbf{g}_m = \sqrt{u_m} \hat{\mathbf{g}}_m$ separates Eq. (D1) exactly into

$$\dot{u}_m = 2u_m(\alpha\lambda_m - \kappa), \quad \dot{\hat{\mathbf{g}}}_m = \alpha(\mathbf{I}_{d \times d} - \hat{\mathbf{g}}_m \hat{\mathbf{g}}_m^\top) \mathbf{C}_r \hat{\mathbf{g}}_m, \quad (\text{D2})$$

with $\lambda_m = \hat{\mathbf{g}}_m^\top \mathbf{C}_r \hat{\mathbf{g}}_m$. The radial equation is Eq. (4). The angular equation is Oja's rule [46]: a power iteration on $\mathbf{C}_r$ constrained to the unit sphere.

Collapse of the angular sector. The angular equation has no m -dependence beyond its initial condition, since every schema is driven by the same matrix $\mathbf{C}_r$. Its stable fixed point is the leading eigenvector of $\mathbf{C}_r$ for all m , so $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n \rightarrow 1$ and $\mathbf{H}^\top \mathbf{H}$ becomes rank one. By Eq. (6) a rank-one repertoire has a single threshold, so the cascade reduces to one bifurcation and the classification of Fig. 2 does not apply. This is the collapse identified for the unweighted rotation $\dot{\mathbf{g}}_m = -\kappa \mathbf{g}_m + \alpha \mathbf{C} \mathbf{g}_m$ ([26, Sec. S1 A]). Resource weighting changes which matrix is iterated but not the fact that one matrix drives every schema.

The collapse is also the opposite of the transformation the theory is meant to describe. Rotation toward the dominant axis of $\mathbf{C}_r$ aligns every schema with the structure that already exists, which reproduces the leading role dimension rather than extending a schema to a context it was not built for.

Repelling mechanisms. Sustaining distinct directions requires a term that separates schemas. Two are available in this model. Neither is contained in Eqs. (1)–(4).

(i) *Imposed lateral inhibition.* Subtracting each schema's overlap with the others,

$$\dot{\mathbf{g}}_m = \alpha(\mathbf{I}_{d \times d} - \hat{\mathbf{g}}_m \hat{\mathbf{g}}_m^\top) \mathbf{C}_r \hat{\mathbf{g}}_m - \eta \sum_{n \neq m} (\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n) \hat{\mathbf{g}}_n, \quad (\text{D3})$$

drives the repertoire onto distinct eigenvectors of $\mathbf{C}_r$ ordered by eigenvalue—the deflation of principal-subspace learning [47]. It is effective but stipulated: the rate η has no counterpart in the payoff structure, so the resulting repertoire is a property of the learning rule rather than of the game.

(ii) *Marginal-return reinforcement.* The redundancy can instead be removed at its source. Coordination income sums over contexts, so two aligned schemas give $\Delta_{i1} = \Delta_{i2}$ and an agent collects W twice for one binary role distinction, while the role information the pair supplies is I rather than $2I$. Additive income therefore violates the Szilard bound of Sec. I whenever schemas are redundant. Charging each schema only its marginal contribution replaces λ_m by

$$\lambda_m^{\text{marg}} = \hat{\mathbf{g}}_m^\top \left(\mathbf{I}_{d \times d} - \sum_{n \neq m} \hat{\mathbf{g}}_n \hat{\mathbf{g}}_n^\top \right) \mathbf{C}_r \hat{\mathbf{g}}_m, \quad (\text{D4})$$

so a schema is reinforced only by variance no other schema already organizes. Rotation then moves toward uncovered variance, which is transposability in Sewell's sense, and the repelling term is derived from the payoff rather than added to the dynamics. The cost is that Eq. (3) and the λ_m of Sec. IV A must both be re-derived, since income is no longer additive across contexts.

We adopt neither mechanism here. The questions this paper asks are ones the strength sector answers on its own.

Appendix E: Regime boundaries

On the principal slice the boundaries are exact analytic curves:

$$\text{budget at capacity: } \Theta = 1, \quad (\text{E1})$$

$$\text{winner-take-all possible: } \Theta = M, \quad (\text{E2})$$

$$\text{Zipf } a = 1: r = 1 \quad (a = 1/r), \quad (\text{E3})$$

$$\text{sat. cluster/geometric: } r = \frac{M}{L(\Theta)}, \quad (\text{E4})$$

$$\text{with } L(\Theta) = \ln \frac{u_{\text{cap}}}{u_0} = \ln \frac{\Theta U_{\text{cap}}}{M u_0}.$$

These curves are exact in the self-averaged ($N_a \rightarrow \infty$) mean-field limit used throughout. Finite populations fluctuate around them, and *which* schema condenses is a symmetry-breaking choice the class does not fix. The map is in this sense a prediction: the sort, its classes,

and these boundaries follow from the near-threshold Gaussian covariance (Sec. IV A) extended into the post-bifurcation thick phase, whose full-loop verification is left to the accompanying simulation program. Boundary Eq. (E4) is where the number of saturated modes $n_{\text{sat}} = \max(0, M - rL)$ reaches zero: below $r = M/L$ a saturated cluster of n_{sat} modes coexists with a geometric tail (a mixed avalanche-plus-log class); above it the cluster is gone. This transition lies within the thick capacity

band, where Fig. 4 marks it by the avalanche and logarithmic sketches rather than a drawn curve. The colour field of Fig. 5a is the bifurcation clustering, defined as the maximal share of bifurcation points $b_k = 1/\mu_k$ falling within a window $\Delta \ln \beta = 0.15$. It is a continuous diagnostic and varies smoothly across these sharp analytic lines. It reaches its floor slightly before Eq. (E4), because a few residual saturated modes still cluster just below $n_{\text{sat}} = 0$.

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- [1] A. Giddens, *The Constitution of Society: Outline of the Theory of Structuration* (Polity Press, Cambridge, UK, 1984).
 - [2] W. H. Sewell, A theory of structure: Duality, agency, and transformation, *American Journal of Sociology* **98**, 1 (1992).
 - [3] A. Acerbi and A. Mesoudi, If we are all cultural darwinians what's the fuss about? clarifying recent disagreements in the field of cultural evolution, *Biology & Philosophy* **30**, 481 (2015).
 - [4] R. E. Lucas Jr, Econometric policy evaluation: A critique, in *Carnegie-Rochester conference series on public policy*, Vol. 1 (North-Holland, 1976) pp. 19–46.
 - [5] R. M. Solow, Building a science of economics for the real world, Testimony before the House Committee on Science and Technology, Subcommittee on Investigations and Oversight (2010), professor Emeritus, MIT.
 - [6] L. Tesfatsion, Agent-based computational economics: A constructive approach to economic theory, in *Handbook of Computational Economics*, Vol. 2, edited by L. Tesfatsion and K. L. Judd (North-Holland, Amsterdam, 2006) Chap. 16, pp. 831–880.
 - [7] J.-P. Bouchaud and M. Mézard, Wealth condensation in a simple model of economy, *Physica A: Statistical Mechanics and its Applications* **282**, 536 (2000).
 - [8] F. C. Santos, J. M. Pacheco, and T. Lenaerts, Cooperation prevails when individuals adjust their social ties, *PLoS Computational Biology* **2**, e140 (2006).
 - [9] A. S. Vezhnevets, J. P. Agapiou, A. Aharon, R. Ziv, J. Matyas, E. A. Duéñez-Guzmán, W. A. Cunningham, S. Osindero, D. Karmon, and J. Z. Leibo, Generative agent-based modeling with actions grounded in physical, social, or digital space using concordia (2023), arXiv:2312.03664 [cs.AI].
 - [10] Anthropic, Patterns and problems in emerging multi-agent systems, <https://www.anthropic.com/research/multiagent-systems> (2026), frontier Red Team.
 - [11] T. A. Han, J. Z. Leibo, T. Lenaerts, I. Rahwan, F. Santos, M. Perc, and V. Capraro, Social physics in the age of artificial intelligence (2026), arXiv:2603.16900 [physics.soc-ph].
 - [12] E. Miehling, K. N. Ramamurthy, K. R. Varshney, M. Riemer, D. Bouneffouf, J. T. Richards, A. Dhurandhar, E. M. Daly, M. Hind, P. Sattigeri, D. Wei, A. Rawat, J. Gajcin, and W. Geyer, Agentic ai needs a systems theory (2025), arXiv:2503.00237 [cs.AI].
 - [13] B. El, J. Paeng, F. Dinc, S. Su, M. Erdogan, A. Pappu, H. Ye, W. Zhao, S. Ganguli, and J. Zou, Physics of agents: Statistical mechanics predicts collective behavior of ai agents (2026), arXiv:2608.16578 [cs.AI].
 - [14] A. Flint, L. M. Aiello, R. Pastor-Satorras, and A. Baronchelli, Group size effects and collective misalignment in llm multi-agent systems, *Proceedings of the National Academy of Sciences* **123**, e2531697123 (2026), <https://www.pnas.org/doi/pdf/10.1073/pnas.2531697123>.
 - [15] J. M. R. Parrondo, J. M. Horowitz, and T. Sagawa, Thermodynamics of information, *Nature Physics* **11**, 131 (2015).
 - [16] L. Peliti and S. Pigolotti, *Stochastic Thermodynamics: An Introduction* (Princeton University Press, Princeton, NJ, 2021).
 - [17] A. P. Castro, J. Bechhoefer, and D. A. Sivak, Harnessing higher-dimensional fluctuations in an information engine, *Phys. Rev. Lett.* **136**, 047101 (2026).
 - [18] L. Szilard, Über die entropieverminderung in einem thermodynamischen system bei eingriffen intelligenter wesen, *Zeitschrift für Physik* **53**, 840 (1929).
 - [19] A. C. Barato and U. Seifert, Stochastic thermodynamics with information reservoirs, *Phys. Rev. E* **90**, 042150 (2014).
 - [20] T. K. Saha, *Information-Powered Engines*, Springer Theses (Springer Nature Switzerland, Cham, 2023).
 - [21] J. M. R. Parrondo, Entropy, macroscopic information, and phase transitions, in *Unsolved Problems of Noise and Fluctuations (UPoN'99)*, AIP Conference Proceedings, Vol. 511 (2000) arXiv:cond-mat/9911066 [cond-mat.stat-mech].
 - [22] B. VanSaders, M. Fruchart, and V. Vitelli, Measurement-induced phase transitions in informational active matter, *PNAS Nexus* **5**, 10.1093/pnasnexus/pgag077 (2026), arXiv:2302.07402 [cond-mat.stat-mech].
 - [23] A. Ziepe, I. Maryshev, I. S. Aranson, and E. Frey, Multi-scale organization in communicating active matter, *Nature Communications* **13**, 6727 (2022).
 - [24] M. I. Jordan, A collectivist, economic perspective on ai (2025), arXiv:2507.06268.
 - [25] N. Tomašev, M. Franklin, J. Jacobs, S. Krier, and S. Osindero, Distributional agi safety (2026), arXiv:2512.16856 [cs.AI].
 - [26] SM, See Supplemental Material for the six-model ladder and the coordination payoff (Sec. S1), the correlated-equilibrium analysis (Sec. S1C), pairing statistics and the non-uniqueness of the Λ factorization (Sec. S2), the associative-memory mapping and role-capacity bound (Sec. S3), and the status-ledger analysis (Sec. S4).
 - [27] M. C. Cross and P. C. Hohenberg, Pattern formation outside of equilibrium, *Rev. Mod. Phys.* **65**, 851 (1993).
 - [28] J. Hofbauer and K. Sigmund, *Evolutionary Games and Population Dynamics* (Cambridge University Press, 1998).

- [29] D. Krotov and J. J. Hopfield, Dense associative memory for pattern recognition, in *Proceedings of the 30th International Conference on Neural Information Processing Systems*, NIPS'16 (Curran Associates Inc., Red Hook, NY, USA, 2016) p. 1180–1188.
- [30] A. M. Saxe, J. L. McClelland, and S. Ganguli, Exact solutions to the nonlinear dynamics of learning in deep linear neural networks, in *International Conference on Learning Representations* (2014) arXiv:1312.6120.
- [31] P. A. Ortega and D. A. Braun, Thermodynamics as a theory of decision-making with information-processing costs, *Proc. R. Soc. A* **469**, 20120683 (2013).
- [32] V. A. Marčenko and L. A. Pastur, Distribution of eigenvalues for some sets of random matrices, *Mathematics of the USSR-Sbornik* **1**, 457 (1967).
- [33] M. Puelma Touzel, The fitness landscape of social norms in social dilemmas (2026), arXiv:2605.18834 [cs.GT].
- [34] H. Gintis, Social norms as choreography, *Politics, Philosophy & Economics* **9**, 251 (2010), <https://doi.org/10.1177/1470594X09345474>.
- [35] H. A. Simon, On a class of skew distribution functions, *Biometrika* **42**, 425 (1955).
- [36] M. Scheffer, J. Bascompte, W. A. Brock, V. Brovkin, S. R. Carpenter, V. Dakos, H. Held, E. H. van Nes, M. Rietkerk, and G. Sugihara, Early-warning signals for critical transitions, *Nature* **461**, 53 (2009).
- [37] P. Ashwin, S. Wiczorek, R. Vitolo, and P. Cox, Tipping points in open systems: bifurcation, noise-induced and rate-dependent examples in the climate system, *Philosophical Transactions of the Royal Society A* **370**, 1166 (2012).
- [38] P. Ritchie and J. Sieber, Early-warning indicators for rate-induced tipping, *Chaos* **26**, 093116 (2016).
- [39] S. J. Gershman, E. J. Horvitz, and J. B. Tenenbaum, Computational rationality: A converging paradigm for intelligence in brains, minds, and machines, *Science* **349**, 273 (2015), <https://www.science.org/doi/pdf/10.1126/science.aac6076>.
- [40] R. Chen, A. Ardit, H. Sleight, O. Evans, and J. Lindsey, Persona vectors: Monitoring and controlling character traits in language models (2025), arXiv:2507.21509 [cs.CL].
- [41] M. Lachmann, S. Számádó, and C. T. Bergstrom, Cost and conflict in animal signals and human language, *Proceedings of the National Academy of Sciences* **98**, 13189 (2001).
- [42] J. M. Horowitz and M. Esposito, Thermodynamics with continuous information flow, *Phys. Rev. X* **4**, 031015 (2014).
- [43] S. Still, Thermodynamic cost and benefit of memory, *Phys. Rev. Lett.* **124**, 050601 (2020).
- [44] D. Daimer and S. Still, Thermodynamically rational decision making under uncertainty (2023), arXiv:2309.10476 [physics.data-an].
- [45] S. Toyabe, T. Sagawa, M. Ueda, E. Muneyuki, and M. Sano, Experimental demonstration of information-to-energy conversion and validation of the generalized Jarzynski equality, *Nature Physics* **6**, 988 (2010).
- [46] E. Oja, A simplified neuron model as a principal component analyzer, *Journal of Mathematical Biology* **15**, 267 (1982).
- [47] T. D. Sanger, Optimal unsupervised learning in a single-layer linear feedforward neural network, *Neural Networks* **2**, 459 (1989).